

## EMS4485

USB Type-C Analog Audio Switch with Protection Function, Moisture and Audio Jack Detection

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# 1. General Description

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The EMS4485 is a high-performance USB Type-C analog switch for the use in portable multimedia devices, supporting analog audio headsets. The EMS4485 can detect OMTP, CTIA, or 3-pole headsets and configure pinouts automatically. The EMS4485 shares common Type-C pins to pass USB 2.0 signals and analog audio signals; the sideband uses wires and analog microphone signals. The EMS4485 also supports high voltage and surge on SBUx pins and USB pins on the USB Type-C receptacle side. The EMS4485 supports the moisture detection for five pins, including CC\_IN, DP\_R, DN\_L, SBU1 and SBU2. Moisture detection function can detect both moisture resistance and voltage. The detection mode includes single-pin mode and multi-pin mode. The multi-pin mode supports the detection polling, and the polling time can be configured through I2C.

# 2. Features and Benefits

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- USB 2.0 switch -3dB bandwidth: -3dB @950 MHz
- Hi-Fi Audio Path: THD+N= -105dB, 1Vrms, f=20~20kHz, 32Ω Load
- Audio Path Insertion loss: -3dB @900 MHz
- Overvoltage protection: DP\_R, DN\_L, SBUx and GSBUX, Vth = 4.4 V (default), 4.4 V ~ 5.0 V configurable
- Support OMTP, CTIA and 3-pole Audio Jack Pinout
- Built-in moisture detection function, including moisture resistor and moisture voltage mode. Support single-pin mode and multi-pin mode. Support polling mode, and the polling time can be configured through I2C.

# 3. Applications

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- USB Type-C Receptacle, Mobile Phone and Tablets

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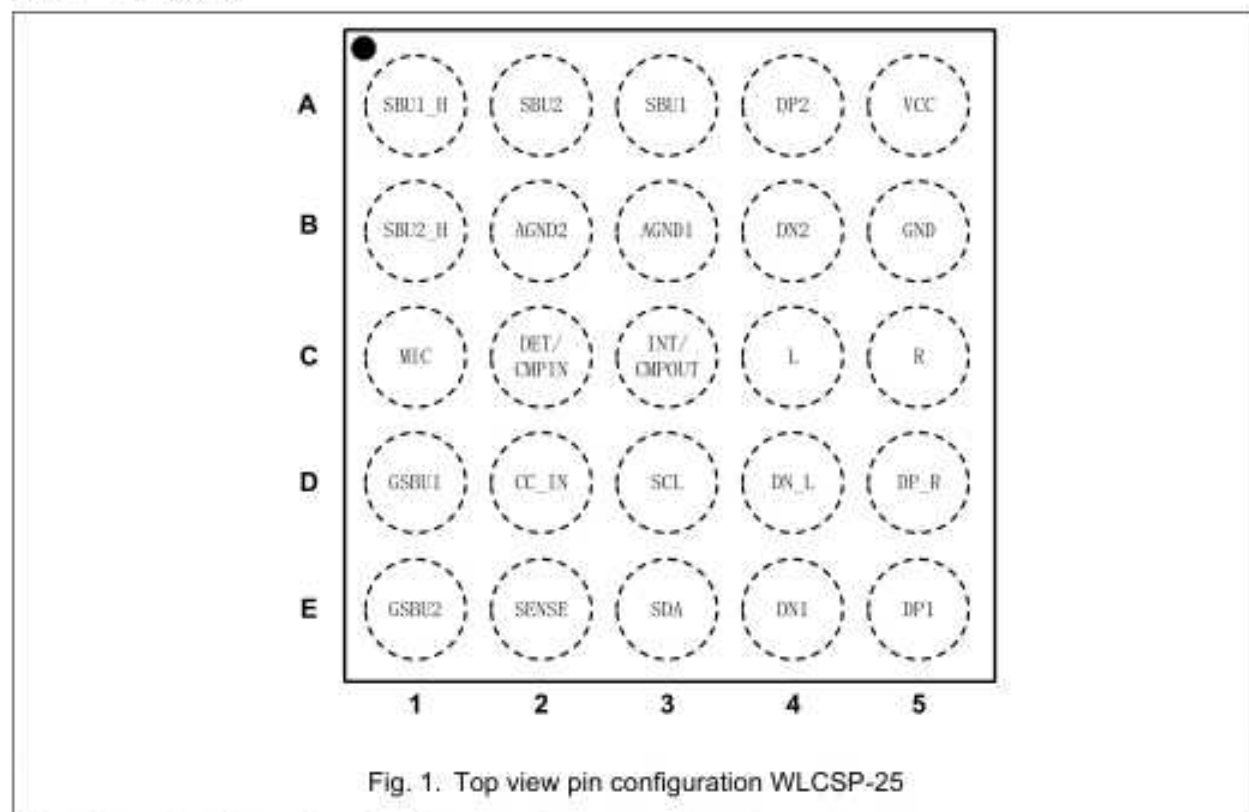
## 4. Ordering Information

Table 1. Ordering information

| Type number | Topside marking | Package  |                                                                                            |          |
|-------------|-----------------|----------|--------------------------------------------------------------------------------------------|----------|
|             |                 | Name     | Description                                                                                | Quantity |
| EMS4485     | A6490           | WLCSP-25 | Wafer-Level Chip Scale Package, 25pins; body width 2 × 2 mm; 0.4 mm pitch, 0.586 mm height | 3000     |

## 5. Pinning Information

### 5.1. Pinning



- (1) B2(AGND2), B3(AGND1), B5(GND) pins must be connected together.
- (2) A3(SBU1), D1(GSBU1) pins must be connected together.
- (3) A2(SBU2), E1(GSBU2) pins must be connected together.

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## 5.2. Pin description

Table 2. Pin description

| Pin No. | Name           | Type | Description                                                                                                          |
|---------|----------------|------|----------------------------------------------------------------------------------------------------------------------|
| A5      | VCC            | PWR  | Positive Supply                                                                                                      |
| B5      | GND            | GND  | Primary Ground Connection. Must be Connected to System Ground                                                        |
| D5      | DP_R           | I/O  | Right Audio / Positive USB2.0 Data Common Line                                                                       |
| D4      | DN_L           | I/O  | Left Audio / Negative USB2.0 Data Common Line                                                                        |
| E5      | DP1            | I/O  | Positive USB2.0 Data Line 1                                                                                          |
| E4      | DN1            | I/O  | Negative USB2.0 Data Line 1                                                                                          |
| C5      | R              | I/O  | Right Line for Audio Signals                                                                                         |
| C4      | L              | I/O  | Left Line for Audio Signals                                                                                          |
| A3      | SBU1           | I/O  | Sideband Use wire 1                                                                                                  |
| A2      | SBU2           | I/O  | Sideband Use wire 2                                                                                                  |
| C1      | MIC            | O    | Analog Audio Microphone                                                                                              |
| B2      | AGND2          | GND  | Analog Ground 2                                                                                                      |
| B3      | AGND1          | GND  | Analog Ground 1                                                                                                      |
| E2      | SENSE          | O    | Analog Ground Sense Return                                                                                           |
| C3      | INT/<br>CMPOUT | O    | Open-drain comparator output, set as CMPOUT when 0x12h bit [4] = 1                                                   |
|         |                |      | Open-drain output, set as INT when 0x12h bit [4] = 0. I2C interrupt output, active low                               |
| D2      | CC_IN          | I    | Audio accessory attachment detection input                                                                           |
| D1      | GSBU1          | I/O  | Star-connection with SBU1 to Headset Jack as Audio Ground Sense Path 1                                               |
| E1      | GSBU2          | I/O  | Star-connection with SBU2 to Headset Jack as Audio Ground Sense Path 2                                               |
| C2      | DET/<br>CMPIN  | O    | Comparator input, set as COMPIN when 0x12h bit [4] = 1                                                               |
|         |                |      | Open-drain output, set as DET when 0x12h bit [4] = 0. When CC_IN > 1.5v, DET is low. When CC_IN < 1.2v, DET is high. |
| D3      | SCL            | I    | I2C Clock wire                                                                                                       |
| E3      | SDA            | I/O  | I2C Data wire                                                                                                        |
| B1      | SBU2_H         | I    | System Side Sideband Use wire 2, can be configured as eDP AUX path                                                   |
| A1      | SBU1_H         | I    | System Side Sideband Use wire 1, can be configured as eDP AUX path                                                   |
| A4      | DP2            | I/O  | Positive USB2.0 Data Line 2                                                                                          |
| B4      | DN2            | I/O  | Negative USB2.0 Data Line 2                                                                                          |

(1) B2(AGND2), B3(AGND1), B5(GND) pins must be connected together.

(2) A3(SBU1), D1(GSBU1) pins must be connected together.

(3) A2(SBU2), E1(GSBU2) pins must be connected together.

## 6. Functional Diagram

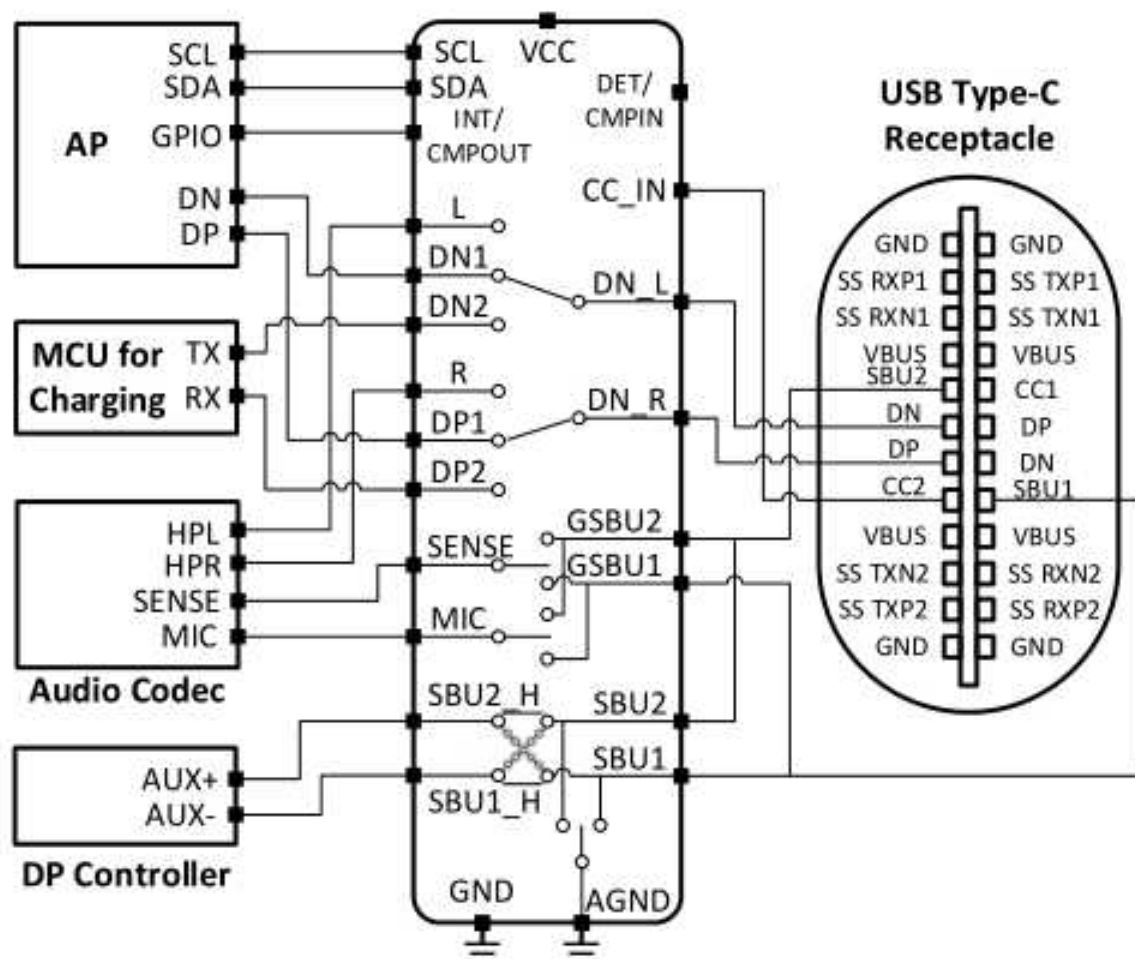


Fig. 2. Functional Diagram

- (1) B2(AGND2), B3(AGND1), B5(GND) pins must be connected together.
- (2) A3(SBU1), D1(GSBUS1) pins must be connected together.
- (3) A2(SBU2), E1(GSBUS2) pins must be connected together.

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## 7. Absolute Maximum Ratings

Stresses beyond those listed in Table-3 Absolute Maximum Ratings may cause permanent damage to the device. They are stress ratings only, which do not imply functional operation of the device at these or any other conditions. Beyond those indicated under Recommended Operating Conditions, exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

**Table 3. Absolute Maximum Ratings**

Over operating free-air temperature range (unless otherwise noted).

Over operating free-air temperature range (unless otherwise noted).

| Symbol                    | Parameter                                                                                                |                       | Min  | Max | Unit |
|---------------------------|----------------------------------------------------------------------------------------------------------|-----------------------|------|-----|------|
| V <sub>CC</sub>           | Supply voltage from V <sub>CC</sub>                                                                      |                       | -0.5 | 5.5 | V    |
| V <sub>CC_IN</sub>        | V <sub>CC_IN</sub> to GND                                                                                |                       | -0.5 | 20  | V    |
| V <sub>SW_C</sub>         | V <sub>OP_R</sub> to GND, V <sub>ON_L</sub> to GND                                                       |                       | -3.3 | 16  | V    |
| V <sub>SW_Audio</sub>     | V <sub>L</sub> to GND, V <sub>R</sub> to GND                                                             |                       | -3.3 | 5.5 | V    |
| V <sub>SW_USB</sub>       | V <sub>DP1</sub> to GND, V <sub>DN1</sub> to GND                                                         |                       | -0.3 | 5.5 | V    |
| V <sub>SW_UART</sub>      | V <sub>DP2</sub> to GND, V <sub>DN2</sub> to GND                                                         |                       | -0.3 | 5.5 | V    |
| V <sub>V_SBUx/GSBUx</sub> | V <sub>SBU1</sub> to GND, V <sub>SBU2</sub> to GND; V <sub>GSBU1</sub> to GND, V <sub>GSBU2</sub> to GND |                       | -0.3 | 16  | V    |
| V <sub>SBUx_H</sub>       | V <sub>SBU1_H</sub> to GND, V <sub>SBU2_H</sub> to GND                                                   |                       | -0.3 | 5.5 | V    |
| V <sub>VD</sub>           | SENSE, MIC to GND                                                                                        |                       | -0.3 | 5.5 | V    |
| V <sub>CTRL</sub>         | Control input voltage                                                                                    | SDA, SCL              | -0.5 | 5.5 | V    |
| V <sub>comparator</sub>   | Comparator input and output                                                                              | DET/CMPIN, INT/CMPOUT | -0.5 | 5.5 | V    |
| I <sub>SW_Audio</sub>     | Switch I/O current, audio path                                                                           |                       | -250 | 250 | mA   |
| I <sub>SW_USB</sub>       | Switch I/O current, USB path                                                                             |                       |      | 100 | mA   |
| I <sub>SW_UART</sub>      | Switch I/O current, UART path                                                                            |                       |      | 50  | mA   |
| I <sub>SW_MIC</sub>       | Switch I/O current, MIC to GSBU1 or GSBU2                                                                |                       |      | 50  | mA   |
| I <sub>SW_SBUx</sub>      | Switch I/O current, SBUx to SBUx_H                                                                       |                       |      | 50  | mA   |
| I <sub>SW_SENSE</sub>     | Switch I/O current, SENSE to GSBU1 or GSBU2                                                              |                       |      | 100 | mA   |
| I <sub>SW_AGND</sub>      | Switch I/O current, AGND to SBU1 or SBU2                                                                 |                       |      | 500 | mA   |
| T <sub>A</sub>            | Absolute maximum operating temperature                                                                   |                       | -40  | 85  | °C   |
| T <sub>STG</sub>          | Storage temperature                                                                                      |                       | -55  | 150 | °C   |

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# 8. Recommended Operating Conditions

Table 4. Recommended Operating Conditions

| Symbol                | Parameter                                                                                            | Min  | Max                                | Unit |
|-----------------------|------------------------------------------------------------------------------------------------------|------|------------------------------------|------|
| V <sub>CC</sub>       | Supply voltage                                                                                       | 3    | 4.5                                | V    |
| V <sub>SW_USB</sub>   | V <sub>DP1</sub> to GND, V <sub>DN1</sub> to GND, V <sub>DP_R</sub> to GND, V <sub>DN_L</sub> to GND | 0    | 5                                  | V    |
| V <sub>SW_UART</sub>  | V <sub>DP2</sub> to GND, V <sub>DN2</sub> to GND, V <sub>DP_R</sub> to GND, V <sub>DN_L</sub> to GND | 0    | V <sub>CC</sub> +0.3               | V    |
| V <sub>SW_Audio</sub> | V <sub>DP_R</sub> to GND, V <sub>DN_L</sub> to GND, V <sub>I</sub> to GND, V <sub>A</sub> to GND     | -3.3 | 5                                  | V    |
| V <sub>VSBU</sub>     | V <sub>SBU1</sub> to GND, V <sub>SBU2</sub> to GND                                                   | 0    | V <sub>CC</sub> +0.3<br>(3.6V Max) | V    |
| V <sub>VCSBU</sub>    | V <sub>CSBU1</sub> to GND, V <sub>CSBU2</sub> to GND                                                 | 0    | 3.6                                | V    |
| V <sub>SEN_MIC</sub>  | V <sub>SENSE</sub> to GND, V <sub>MIC</sub> to GND                                                   | 0    | 5                                  | V    |
| V <sub>VSBU_H</sub>   | V <sub>SBU1_H</sub> to GND, V <sub>SBU2_H</sub> to GND                                               | 0    | 5                                  | V    |
| V <sub>CC_IN</sub>    | V <sub>CC_IN</sub> to GND                                                                            | 0    | 18                                 | V    |
| V <sub>IH</sub>       | Input voltage high                                                                                   | 1.3  | V <sub>CC</sub> +0.3               | V    |
| V <sub>IL</sub>       | Input voltage low                                                                                    |      | 0.3                                | V    |
| T <sub>A</sub>        | Ambient operating temperature                                                                        | -40  | 85                                 | °C   |

1. The input and output negative voltage ratings may be exceeded if the input and output diode current ratings are observed.
2. Control input must be held high or Low, it must not float.
3. In USB mode, any signal applied to the off-state audio inputs R, L may not swing below ground or above 1V.

# 9. ESD Ratings

When a statically-charged person or object touches an electrostatic discharge sensitive device, the electrostatic charge might be drained through sensitive circuitry in the device. If the electrostatic discharge possesses sufficient energy, damage might occur to the device due to localized overheating.

Table 5. ESD Ratings

| Parameter   | Conditions                       | Value | Unit |
|-------------|----------------------------------|-------|------|
| ESD voltage | HBM, DP_R, DN_L, SENSE, MIC pins | ±5    | kV   |
|             | HBM, all other pins              | ±2.5  | kV   |
|             | Latch-up                         | 800   | mA   |

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## 10. Electrical Characteristics

### 10.1. DC electrical characteristics

The values are obtained under these conditions unless otherwise specified:  $V_{CC} = 2.7\text{ V to }4.5\text{ V}$ ,  $V_{CC} = 3.3\text{ V (Typ.)}$ ,  $T_A = -40^\circ\text{C to }85^\circ\text{C}$ , and  $T_A = 25^\circ\text{C (Typ.)}$

| Symbol                            | Parameter                                              | Conditions                                               | Power                            | Min | Typ | Max | Unit |
|-----------------------------------|--------------------------------------------------------|----------------------------------------------------------|----------------------------------|-----|-----|-----|------|
| I <sub>CC</sub>                   | Supply current                                         | USB switches on, SBUx to SBUx_H switches on              | V <sub>CC</sub> = 4.2 V          |     | 55  |     | μA   |
|                                   |                                                        | Audio switches on, MIC switch on and Audio GND switch on |                                  |     | 68  |     | μA   |
| I <sub>CCZ</sub>                  | Standby current                                        | 04H'b7 = 0                                               |                                  |     |     | 5   |      |
| USB/Audio common pins: DP_R, DN_L |                                                        |                                                          |                                  |     |     |     |      |
| I <sub>OZ</sub>                   | Off leakage current of DP_R and DN_L                   | DN_L, DP_R = 0 V to 3.6 V                                | V <sub>CC</sub> = 2.7 V to 4.5 V | -3  |     | 3   | μA   |
| I <sub>OFF</sub>                  | Power-off leakage current of DP_R and DN_L             | DN_L, DP_R = 0 V to 3.6 V                                | Power off                        | -3  |     | 3   | μA   |
| V <sub>OV_TRIP</sub>              | Input OVP lockout                                      | Rising edge, 0x12h bit [7:6] = 00                        | V <sub>CC</sub> = 2.7 V to 4.5 V |     | 4.4 |     | V    |
| V <sub>OV_HYS</sub>               | OVP hysteresis                                         |                                                          |                                  |     | 0.4 |     | V    |
| Audio switch                      |                                                        |                                                          |                                  |     |     |     |      |
| I <sub>ON</sub>                   | On leakage current of audio switch                     | DN_L, DP_R = -3 V to 3 V, DP, DN, R, L = Float           | V <sub>CC</sub> = 2.7 V to 4.5 V | -5  |     | 5   | μA   |
| R <sub>ON</sub>                   | Switch on resistance                                   | I <sub>SW</sub> = 100 mA, V <sub>SW</sub> = -3 V to 3 V  | V <sub>CC</sub> = 2.7 V to 4.5 V |     | 1   |     | Ω    |
| R <sub>SHUNT</sub>                | Pull-down resistor on R/L pin when audio switch is off | L = R = 3 V                                              |                                  |     |     | 10  |      |
| USB switch                        |                                                        |                                                          |                                  |     |     |     |      |

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|                     |                                          |                                                          |                                            |    |     |   |               |
|---------------------|------------------------------------------|----------------------------------------------------------|--------------------------------------------|----|-----|---|---------------|
| $I_{ON}$            | On leakage current of USB switch         | DN_L, DP_R = 0 V to 3.6 V,<br>DP1, DN1, R, L = Float     | $V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$ | -3 |     | 3 | $\mu\text{A}$ |
| $I_{OZ}$            | Off leakage current of DP1 and DN1       | DP1, DN1 = 0 V to 3.6 V                                  | $V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$ | -3 |     | 3 | $\mu\text{A}$ |
| $I_{OFF}$           | Power-off leakage current of DP1 and DN1 | DP1, DN1 = 0 V to 3.6 V                                  | Power off                                  | -3 |     | 3 | $\mu\text{A}$ |
| $R_{ON\_USB}$       | USB switch on resistance                 | $I_{SW} = 8 \text{ mA}$ ,<br>$V_{SW} = 0.4 \text{ V}$    | $V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$ |    | 6   |   | $\Omega$      |
| <b>UART switch</b>  |                                          |                                                          |                                            |    |     |   |               |
| $I_{OZ}$            | Off leakage current of DP2 and DN2       | DP2, DN2 = 0 V to 3.6 V                                  | $V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$ | -3 |     | 3 | $\mu\text{A}$ |
| $R_{ON\_UART}$      | UART switch on resistance                | $I_{SW} = 3 \text{ mA}$ ,<br>$V_{SW} = 0.4 \text{ V}$    | $V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$ |    | 17  |   | $\Omega$      |
| $I_{ON}$            | On leakage current UART switch           | DN_L, DP_R = 0 V to 3.6 V,<br>DP2, DN2, R, L = Float     | $V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$ | -3 |     | 3 | $\mu\text{A}$ |
| <b>SENSE switch</b> |                                          |                                                          |                                            |    |     |   |               |
| $I_{ON}$            | SENSE path leakage current               | GSBUx = 0 V to 1 V,<br>SENSE is floating                 | $V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$ | -2 |     | 2 | $\mu\text{A}$ |
| $R_{ON}$            | SENSE switch on resistance               | $I_{OUT} = 100 \text{ mA}$ ,<br>$V_{SW} = 1.0 \text{ V}$ | $V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$ |    | 1.2 |   | $\Omega$      |
| $I_{OFF}$           | Power-off leakage current of SENSE       | SENSE = 0 V to 1.0 V                                     | Power off                                  | -2 |     | 2 | $\mu\text{A}$ |
|                     | Power-off leakage current of GSBUx       | GSBUx = 0 V to 3.6 V                                     |                                            | -3 |     | 3 | $\mu\text{A}$ |
| $V_{OV\_TRIP}$      | Input OVP lockout on GSBUx               | Rising edge,<br>0x12h bit [7:6] = 00                     | $V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$ |    | 4.4 |   | V             |

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|                                       |                                     |                                                                        |                                            |    |     |   |               |
|---------------------------------------|-------------------------------------|------------------------------------------------------------------------|--------------------------------------------|----|-----|---|---------------|
| $V_{OV\_HYS}$                         | Input OVP hysteresis of GSBUX       |                                                                        |                                            |    | 0.4 |   | V             |
| SBUx pins                             |                                     |                                                                        |                                            |    |     |   |               |
| $I_{OZ}$                              | Off leakage current of SBUx         | SBUx = 0 V to 3.6 V                                                    | $V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$ | -3 |     | 3 | $\mu\text{A}$ |
| $I_{OFF}$                             | Power-off leakage current port SBUx | SBUx = 0 V to 3.6 V                                                    | Power off                                  | -3 |     | 3 | $\mu\text{A}$ |
| $V_{OV\_TRIP}$                        | Input OVP lockout                   | Rising edge, 0x12h bit [7:6] = 00                                      | $V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$ |    | 4.4 |   | V             |
| $V_{OV\_HYS}$                         | Input OVP hysteresis                |                                                                        |                                            |    | 0.4 |   | V             |
| MIC switch                            |                                     |                                                                        |                                            |    |     |   |               |
| $I_{ON}$                              | On leakage current of MIC switch    | SBUx = 0 V to 3.6 V, MIC is floating                                   | $V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$ | -3 |     | 3 | $\mu\text{A}$ |
| $I_{OFF}$                             | Power-off leakage current of MIC    | MIC = 0 V to 3.6 V                                                     | Power off                                  | -1 |     | 1 | $\mu\text{A}$ |
| $R_{ON}$                              | MIC switch on resistance            | $V_{SW} = 1\text{V}$ ,<br>$I_{SW} = 30 \text{ mA}$                     | $V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$ |    | 11  |   | $\Omega$      |
| SBUx_H switch                         |                                     |                                                                        |                                            |    |     |   |               |
| $I_{ON}$                              | On leakage current of SBUx_H switch | SBUx = 0 V to 3.6 V, SBUx_H is floating                                | $V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$ | -3 |     | 3 | $\mu\text{A}$ |
| $I_{OFF}$                             | Power off leakage current of SBUx_H | SBUx_H = 0 V to 3.6 V                                                  | Power off                                  | -1 |     | 1 | $\mu\text{A}$ |
| $R_{ON}$                              | SBUx_H switch on resistance         | $V_{SW} = 0 \text{ V to } 3.6 \text{ V}$ ,<br>$I_{SW} = 30 \text{ mA}$ | $V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$ |    | 10  |   | $\Omega$      |
| Audio ground switch pin: AGND to SBUx |                                     |                                                                        |                                            |    |     |   |               |
| $R_{ON}$                              | AGND switch on resistance           | $I_{SOURCE} = 100 \text{ mA}$ on SBUx                                  | $V_{CC} = 2.7 \text{ V to } 4.5 \text{ V}$ |    | 150 |   | m $\Omega$    |
| CC_IN pin                             |                                     |                                                                        |                                            |    |     |   |               |

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|               |                                   |                        |                      |     |     |     |         |
|---------------|-----------------------------------|------------------------|----------------------|-----|-----|-----|---------|
| $V_{TH\_L}$   | Input low threshold               |                        | VCC = 2.7 V to 4.5 V |     | 1.2 |     | V       |
| $V_{TH\_H}$   | Input high threshold              |                        |                      |     | 1.5 |     | V       |
| $I_{IN}$      | Input leakage of CC_IN            | CC_IN = 0 V to 5.5 V   |                      | -3  |     | 1   | $\mu$ A |
| SDA, SCL pins |                                   |                        |                      |     |     |     |         |
| $V_{IL2C}$    | Low-level input voltage           |                        | VCC = 2.7 V to 4.5 V |     |     | 0.3 | V       |
| $V_{IH2C}$    | High-level input voltage          |                        |                      | 1.3 |     |     | V       |
| $I_{I2C}$     | Input current of SDA and SCL pins | SCL/SDA = 0 V to 3.6 V |                      | -5  |     | 5   | $\mu$ A |
| $V_{OLSDA}$   | Low-level output voltage          | $I_{OL} = 2$ mA        |                      |     |     | 0.3 | V       |
| $I_{OLSDA}$   | Low-level output current          | $V_{OLSDA} = 0.2$ V    |                      | 10  |     |     | mA      |

**Note:**

- (4) Flatness is defined as the difference between maximum and minimum value of ON-resistance at the specified analog signal voltage points.
- (5) RON matching between channels is calculated by subtracting the channel with the lowest max Ron value from the channel with the highest max Ron value.
- (6) The OVP voltage can be adjusted by bit [7:6] for 12h: 4.4 V, 4.6 V, 4.8 V and 5 V, each step is 0.2V.
- (7) Specifications subject to change without notice.

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### USB Type-C Analog Audio Switch with Protection Function, Moisture and Audio Jack Detection

## 10.2. AC electrical characteristics

The values are obtained under these conditions unless otherwise specified: VCC = 2.7 V to 4.5 V, VCC = 3.3 V (Typ.), TA = -40°C to 85°C, and TA = 25°C (Typ.)

| Symbol                           | Parameter                                                             | Conditions                                                                                       | Power       | Min | Typ  | Max | Unit |
|----------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------|-----|------|-----|------|
| Audio switch                     |                                                                       |                                                                                                  |             |     |      |     |      |
| t <sub>delay</sub>               | Audio switch turn-on delay time                                       | DP_R = DN_L = 1 V,<br>R <sub>L</sub> = 32 Ω                                                      | VCC = 3.3 V |     | 410  |     | μs   |
| t <sub>rise</sub>                | Audio switch turn-on rising time                                      |                                                                                                  |             |     | 230  |     | μs   |
| t <sub>OFF</sub>                 | Audio switch turn-off time                                            |                                                                                                  |             |     | 45   |     | ns   |
| X <sub>TALK</sub> <sup>(1)</sup> | Cross talk (adjacent)                                                 | f = 1 kHz, R <sub>L</sub> = 50 Ω,<br>V <sub>SW</sub> = 1 V <sub>RMS</sub>                        | VCC = 3.3 V |     | -88  |     | dB   |
| BW                               | -3 dB bandwidth                                                       | R <sub>L</sub> = 50 Ω                                                                            |             |     | 900  |     | MHz  |
| O <sub>IRR</sub>                 | Off isolation                                                         | f = 1 kHz, R <sub>L</sub> = 50 Ω,<br>C <sub>L</sub> = 0 pF, V <sub>SW</sub> = 1 V <sub>RMS</sub> |             |     | -108 |     | dB   |
| THD+N                            | Total harmonic distortion + noise performance with A-weighting filter | R <sub>L</sub> = 600 Ω,<br>f = 20 Hz ~ 20 kHz,<br>V <sub>SW</sub> = 2 V <sub>RMS</sub>           |             |     | -108 |     | dB   |
|                                  |                                                                       | R <sub>L</sub> = 32 Ω,<br>f = 20 Hz ~ 20 kHz,<br>V <sub>SW</sub> = 1 V <sub>RMS</sub>            |             |     | -105 |     | dB   |
|                                  |                                                                       | R <sub>L</sub> = 16 Ω,<br>f = 20 Hz ~ 20 kHz,<br>V <sub>SW</sub> = 0.5 V <sub>RMS</sub>          |             |     | -101 |     | dB   |
| USB switch                       |                                                                       |                                                                                                  |             |     |      |     |      |
| t <sub>ON</sub>                  | USB switch turn-on time                                               | DP_R = DN_L = 1.5 V,<br>R <sub>L</sub> = 50 Ω                                                    | VCC = 3.3 V |     | 120  |     | ns   |
| t <sub>OFF</sub>                 | USB switch turn-off time                                              |                                                                                                  |             |     | 13   |     | ns   |
| BW                               | -3 dB bandwidth                                                       | R <sub>L</sub> = 50 Ω                                                                            |             |     | 950  |     | MHz  |
| O <sub>IRR</sub>                 | Off isolation between DP, DN and common node pins                     | f = 1 kHz, R <sub>L</sub> = 50 Ω,<br>C <sub>L</sub> = 0 pF, V <sub>SW</sub> = 1 V <sub>RMS</sub> |             |     | -107 |     | dB   |

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|                                  |                                      |                                                                     |                       |  |      |  |               |
|----------------------------------|--------------------------------------|---------------------------------------------------------------------|-----------------------|--|------|--|---------------|
| $t_{OVP}$                        | DP_R and DN_L pins OVP response time | $V_{SW} = 3.5 \text{ V to } 5.5 \text{ V}$                          |                       |  | 1    |  | $\mu\text{s}$ |
| <b>UART switch</b>               |                                      |                                                                     |                       |  |      |  |               |
| $t_{ON}$                         | UART switch turn-on time             | $DP\_R = DN\_L = 1.5 \text{ V}, R_L = 50 \Omega$                    | $VCC = 3.3 \text{ V}$ |  | 90   |  | ns            |
| $t_{OFF}$                        | UART switch turn-off time            |                                                                     |                       |  | 6    |  | ns            |
| BW                               | -3 dB bandwidth                      | $R_L = 50 \Omega$                                                   |                       |  | 720  |  | MHz           |
| <b>MIC / Audio ground switch</b> |                                      |                                                                     |                       |  |      |  |               |
| $t_{delay\_MIC}$                 | MIC switch turn-on delay time        | $SBUx = 1 \text{ V}, R_L = 50 \Omega$                               | $VCC = 3.3 \text{ V}$ |  | 1.75 |  | ms            |
| $t_{rise\_MIC}$                  | MIC switch turn-on rising time       |                                                                     |                       |  | 1.5  |  | ms            |
| $t_{OFF\_MIC}$                   | MIC switch turn-off time             |                                                                     |                       |  | 30   |  | ns            |
| $t_{delay\_AGND}$                | AGND switch turn-on time             | $SBUx$ pulled to 0.2 V by 8 $\Omega$ , AGND connect to GND          | $VCC = 3.3 \text{ V}$ |  | 430  |  | $\mu\text{s}$ |
| $t_{rise\_AGND}$                 | AGND switch turn-on rising time      |                                                                     |                       |  | 280  |  | $\mu\text{s}$ |
| $t_{OFF\_AGND}$                  | AGND switch turn-off time            | $SBUx: I_{SOURCE} = 10 \text{ mA}, \text{ clamp to } 2.5 \text{ V}$ |                       |  | 140  |  | ns            |
| BW                               | MIC switch bandwidth                 | $R_L = 50 \Omega$                                                   |                       |  | 55   |  | MHz           |
| <b>SBUx_H switch</b>             |                                      |                                                                     |                       |  |      |  |               |
| $t_{ON}$                         | SBUx_H switch turn-on time           | $SBUx = 2.5 \text{ V}, R_L = 50 \Omega$                             | $VCC = 3.3 \text{ V}$ |  | 320  |  | $\mu\text{s}$ |
| $t_{OFF}$                        | SBUx_H switch turn-off time          |                                                                     |                       |  | 12   |  | ns            |
| BW                               | Bandwidth                            | $R_L = 50 \Omega$                                                   |                       |  | 45   |  | MHz           |
| $t_{OVP}$                        | SBUx pins OVP response time          | $V_{SW} = 3.5 \text{ V to } 5.5 \text{ V}$                          | $VCC = 3.3 \text{ V}$ |  | 0.8  |  | $\mu\text{s}$ |

**Note:**

(1) Crosstalk is inversely proportional to source impedance.

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## 10.3. Comparator electrical characteristics

The values are obtained under these conditions unless otherwise specified: VCC = 2.7 V to 4.5 V, VCC = 3.3 V (Typ.) and TA = 25°C (Typ.)

| Symbol                     | Parameter                 | Conditions           | Min  | Typ | Max       | Unit |
|----------------------------|---------------------------|----------------------|------|-----|-----------|------|
| <b>Offset voltage</b>      |                           |                      |      |     |           |      |
| V <sub>REF</sub>           | Reference voltage         | VCC = 2.7 V to 4.5 V |      | 216 |           | mV   |
| V <sub>HYS</sub>           | Hysteresis                | VCC = 2.7 V to 4.5 V |      | 20  |           | mV   |
| <b>Input voltage range</b> |                           |                      |      |     |           |      |
| V <sub>CM</sub>            | Common-mode voltage range | VCC = 2.7 V to 4.5 V | -0.1 |     | VCC + 0.1 | V    |

**Note:**

(1) Guaranteed by design.

## 10.4. Capacitance

The values are obtained under these conditions unless otherwise specified: VCC = 3.3 V (Typ.) and TA = 25°C (Typ.)

| Symbol                         | Parameter                          | Conditions                                          | Min | Typ | Max | Unit |
|--------------------------------|------------------------------------|-----------------------------------------------------|-----|-----|-----|------|
| C <sub>ON_USB/Audio</sub>      | On capacitance (common port)       | f = 1 MHz, 100 mV <sub>PK-PK</sub> , 100 mV DC bias |     | 22  |     | pF   |
| C <sub>OFF_USB/Audio</sub>     | Off capacitance (common port)      | f = 1 MHz, 100 mV <sub>PK-PK</sub> , 100 mV DC bias |     | 5.6 |     | pF   |
| C <sub>OFF_USB</sub>           | Off capacitance (non-common ports) | f = 1 MHz, 100 mV <sub>PK-PK</sub> , 100 mV DC bias |     | 0.5 |     | pF   |
| C <sub>ON_SENSE_SW</sub>       | On capacitance (common port)       | f = 1 MHz, 100 mV <sub>PK-PK</sub> , 100 mV DC bias |     | 38  |     | pF   |
| C <sub>ON_MIC_SW</sub>         | On capacitance (common port)       | f = 1 MHz, 100 mV <sub>PK-PK</sub> , 100 mV DC bias |     | 25  |     | pF   |
| C <sub>OFF_SENSE /MIC_SW</sub> | Off capacitance (common port)      | f = 1 MHz, 100 mV <sub>PK-PK</sub> , 100 mV DC bias |     | 6.8 |     | pF   |
| C <sub>ON_SBUx_H_SW</sub>      | On capacitance (common port)       | f = 1 MHz, 100 mV <sub>PK-PK</sub> , 100 mV DC bias |     | 59  |     | pF   |
| C <sub>OFF_SBUx_H_SW</sub>     | Off capacitance (common port)      | f = 1 MHz, 100 mV <sub>PK-PK</sub> , 100 mV DC bias |     | 40  |     | pF   |

**Note:**

(1) Guaranteed by design.

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### USB Type-C Analog Audio Switch with Protection Function, Moisture and Audio Jack Detection

## 10.5. I2C specification

### 10.5.1. I2C Controlling

EMS4485 switching functions are controlled by 2 I2C pins: SCL and SDA pin. The timing characteristics and diagrams, as well as internal register meaning and usage are listed below:

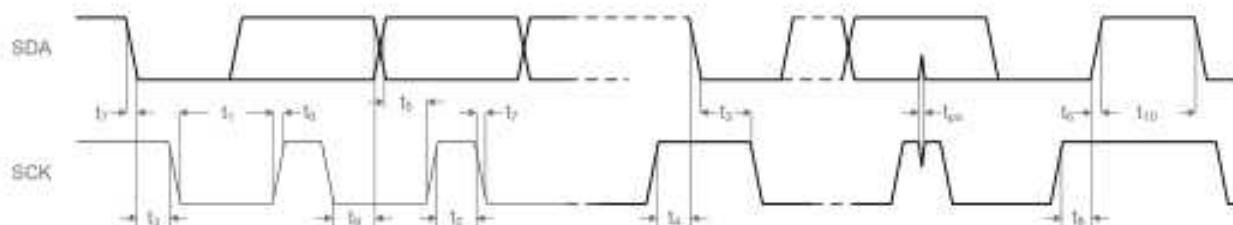


Fig. 3. Definition of timing for full-speed mode devices on the I2C bus

**Note:** Each of SDA and SCL pins should be pulled up by a 2.2kΩ resistor.

### 10.5.2. I2C Timing Characteristics

The values are obtained under these conditions unless otherwise specified: VCC = 3.3 V (Typ.) and TA = 25°C (Typ.)

| Symbol    | Parameter                                       | Conditions | Min | Typ | Max | Unit |
|-----------|-------------------------------------------------|------------|-----|-----|-----|------|
| $f_{CLX}$ | SCL Frequency                                   |            |     |     | 400 | kHz  |
| $t_1$     | SCL Low Pulse-Width                             |            | 1.3 |     |     | μs   |
| $t_2$     | SCL High Pulse-Width                            |            | 0.6 |     |     | μs   |
| $t_3$     | Hold Time (Start Condition)                     |            | 0.6 |     |     | μs   |
| $t_4$     | Setup Time (Start Condition)                    |            | 0.6 |     |     | μs   |
| $t_5$     | Data setup time                                 |            | 0.1 |     |     | μs   |
| $t_6$     | SDA, SCL Rise Time                              |            |     |     | 0.3 | μs   |
| $t_7$     | SDA, SCL Fall Time                              |            |     |     | 0.3 | μs   |
| $t_8$     | Setup Time (Stop Condition)                     |            | 0.6 |     |     | μs   |
| $t_9$     | Data Hold Time                                  |            |     |     | 0.9 | μs   |
| $t_{10}$  | Bus-free time between Stop and Start conditions |            | 1.3 |     |     | μs   |
| $t_{ps}$  | Pulse Width of Spikes being Suppressed          |            |     |     | 50  | ns   |

## 11. Application Information

### 11.1. Overvoltage protection

The EMS4485 features an over voltage protection (OVP) on receptacle side pins that turns off the internal signal routing path if the voltage exceeds the OVP threshold. If an OVP is occurred, flag register 0x02h and 0x03h will indicate which pin had the OVP event. OVP threshold voltage is configurable by 0x12h, bit [7:6].

### 11.2. Watchdog timer

The function of watchdog is to continuously monitor I2C communication. A timer will start as long as the watchdog is enabled. The watchdog receives either read or write command from the master to be reset with I2C communication. In case there's no I2C communication detected after the timer counting down to zero, EMS4485 will control Register 0x04h, 0x05h and 0x12h bit[1] to their default value, reset switch setting and moisture detection enable/disable signals. Watchdog timer can be programmed through Register 0x20h.

### 11.3. Headset detection

The EMS4485 integrates headset unplug detection function by detecting the CC\_IN voltage. The function will be active when device is enabled. Register 0x11h, bit [1:0] Output can indicate if CC\_IN is low ( $CC\_IN < 1.2\text{ V}$ ) or high ( $CC\_IN > 1.5\text{ V}$ ).

|                         | 0x11h, bit [1] | 0x11h, bit [0] |
|-------------------------|----------------|----------------|
| $CC\_IN < 1.2\text{ V}$ | 0              | 1              |
| $CC\_IN > 1.5\text{ V}$ | 1              | 0              |

### 11.4. MIC switch auto-off function

The function is active during control bit 0x12h bit [2] = 1. When CC\_IN is changed from low to high, and L, R, and AGND switches are on, the MIC switch will be off.

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#### 11.5. Audio jack detection and configuration

The function is active when the headset is inserted and CC\_IN is changed from high to low. The EMS4485 can detect OMTP, CTIA or 3-pole headset and configure pinout automatically. During detection and configuration, the R, L, Sense, MIC and Audio ground switches will be off. After detection and configuration, R, L, MIC, Sense and AGND switches will turn on according to detection results and timing control setting.

#### 11.6. I<sup>2</sup>C interface

The EMS4485 includes a full I<sup>2</sup>C slave controller. The I<sup>2</sup>C slave fully complies with the I<sup>2</sup>C specification version 2.1 requirements. This block is designed for fast mode, 400 kHz, signals. Examples of an I<sup>2</sup>C write and read sequence are shown in below figures respectively.

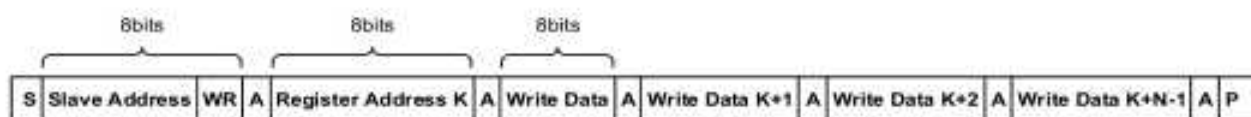


Fig. 4. I<sup>2</sup>C write example

Note: Single-byte read is initiated by the master with P immediately following the first data byte.



Fig. 5. I<sup>2</sup>C read example

Note: If the register is not specified, the master will begin reading from the current register. In this case only sequence showing in the red bracket is needed.

#### 11.7. Moisture detection

The EMS4485 can detect moisture in type-C connector. When moisture detection is enabled, all the related channels will be closed. The voltage can be measured from the selected pin with current source enabled or disabled according to 0x13h bit [3]. Leakage between the selected pins and GND/VBUS can be detected by comparing the measured results with pre-set thresholds, and

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it will pull down "INT", when moisture is found. The related channels will be recovered when the pin detection is completed.

An internal current source will be applied to the pin first during detection of the leakage between GND and pin CC\_IN, DP\_R, DN\_L, SBU1 or SBU2. After getting the result reflecting the resistance from this pin to GND, the result will be stored to corresponding register.

The voltage from the pin without current source will be measured during detection of the leakage between VBUS and pin CC\_IN, DP\_R, DN\_L, SBU1 or SBU2. For the leakage from VBUS and the voltage drop from this pin to ground, they will be detected by the measurement when the moisture detection is performed. The moisture detection has single-pin mode and multi-pin mode.

#### Single-pin mode

In single-pin mode, the moisture detection result would be stored separately in register 0x14h (with current source enabled) or 0x30h (with current source disabled).

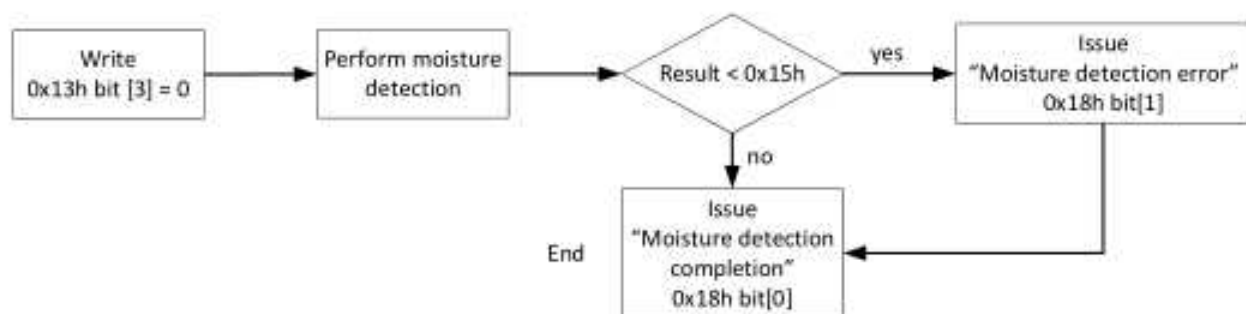


Fig. 6. Single-pin mode (with current source enabled)

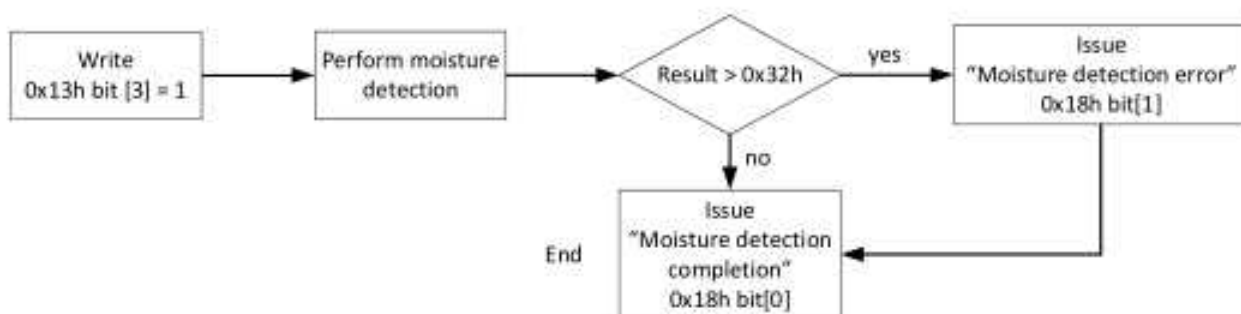


Fig. 7. Single-pin mode (with current source disabled)

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#### Multi-pin mode

In multi-pin mode, if all pins are selected, the turn of moisture detecting sequence will be CC\_IN, DP\_R, DN\_L, SBU1, SBU2. In case some pins are not selected, the detection will ignore the unselected pins and performed on the next selected pins. For example, if the pin CC\_IN, DP\_L are not selected, the moisture detection will be performed on DP\_R first, then SBU1 and SBU2 in turn.

As soon as the detection starts, whether the setup of the current source or pin selection would be ignored. And all the related channels would be forced off during the whole round detection (from the 1st pin to the last pin). The related channels will be recovered after the detection for all pins are completed.

If moisture error is detected in some pins (result > 0x32h for detection with current source disabled, or the result < 0x15h with current source enabled), EMS4485 will complete the whole round, and then stop moisture detection and indicate the flag of "Moisture detection completion". 0x31h bit [5] will select the blank time between two adjacent detections. It's shown in fig 6, fig 7 as below.

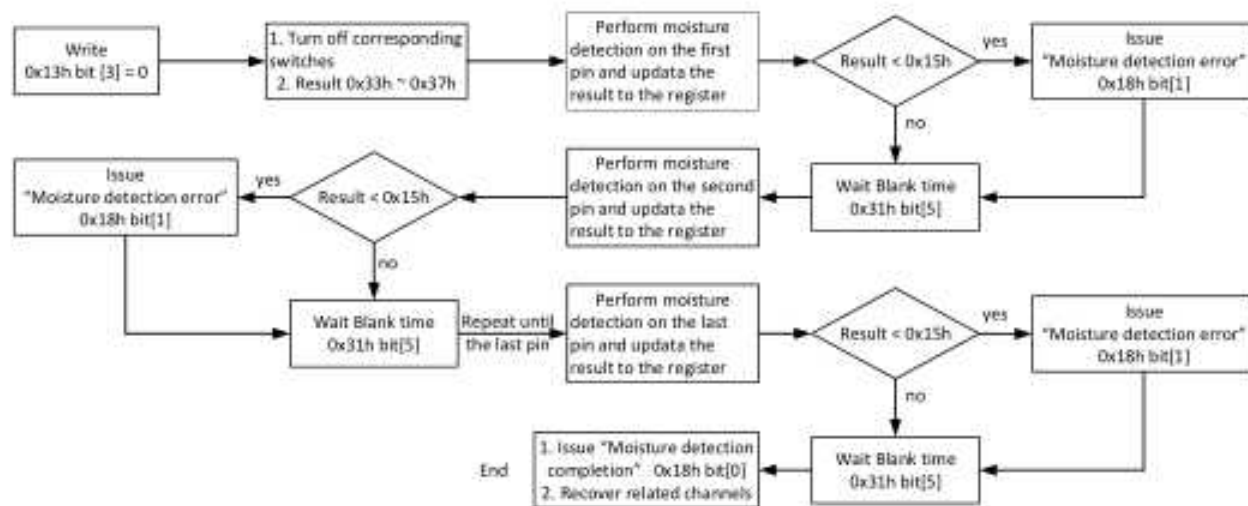


Fig. 8. Multi-pin mode (with current source enabled)

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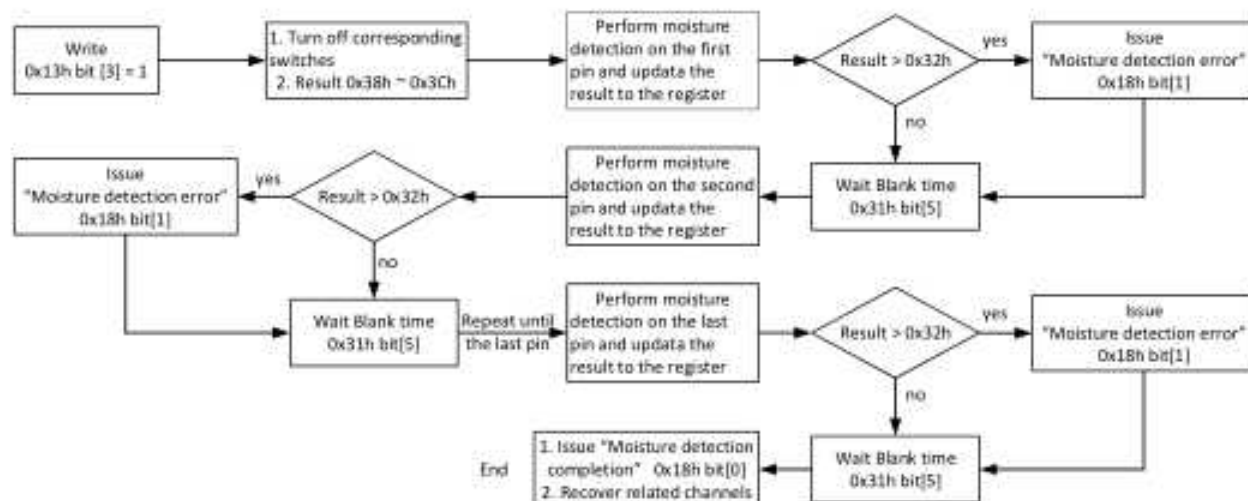


Fig. 9. Multi-pin mode (with current source disabled)

#### Polling

With 0x16h bit [1:0] = 1, EMS4485 can also repeatedly perform moisture detection on its own. A list of scenarios with the combination set of 0x16h bit [1:0] is shown as below. In single-pin mode, when there's no moisture detected (result <= 0x32h for detection with current source disabled, or result >= 0x15h with current source enabled), the moisture detection will be restarted after a period set by 0x16h bit [1:0].

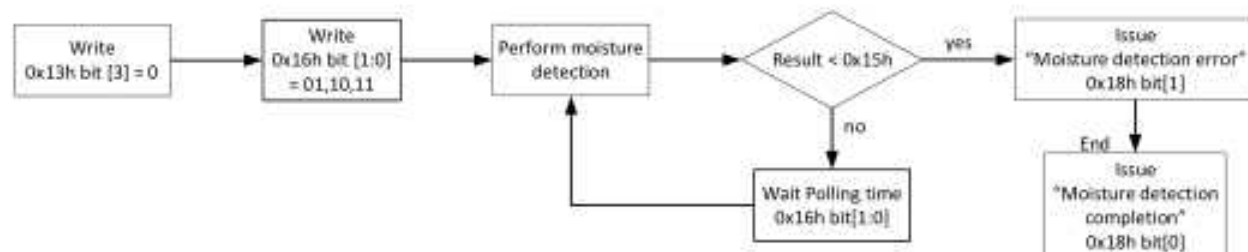


Fig. 10. Single-pin polling mode (with current source enabled)

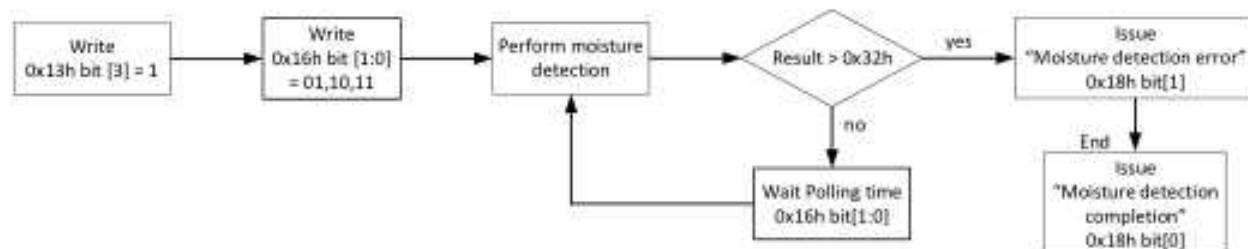


Fig. 11. Single-pin polling mode (with current source disabled)

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In multi-pin mode, when the measured result is ok after one round detection for selected pins, a new round will restart after the period set by Polling time. In the polling period, any attempt of changing the setup of current source or selecting the pins will be noticed in preparation of the next new round.

During the detection, all the related channels will be closed as soon as the first pin detection starts and the channels will not recover until the last pin detection finishes, which means this round detection is completed. To exit polling immediately, there are two situations: one situation is writing 0x13h bit [7] = 1, the other one is an OVP event detected.

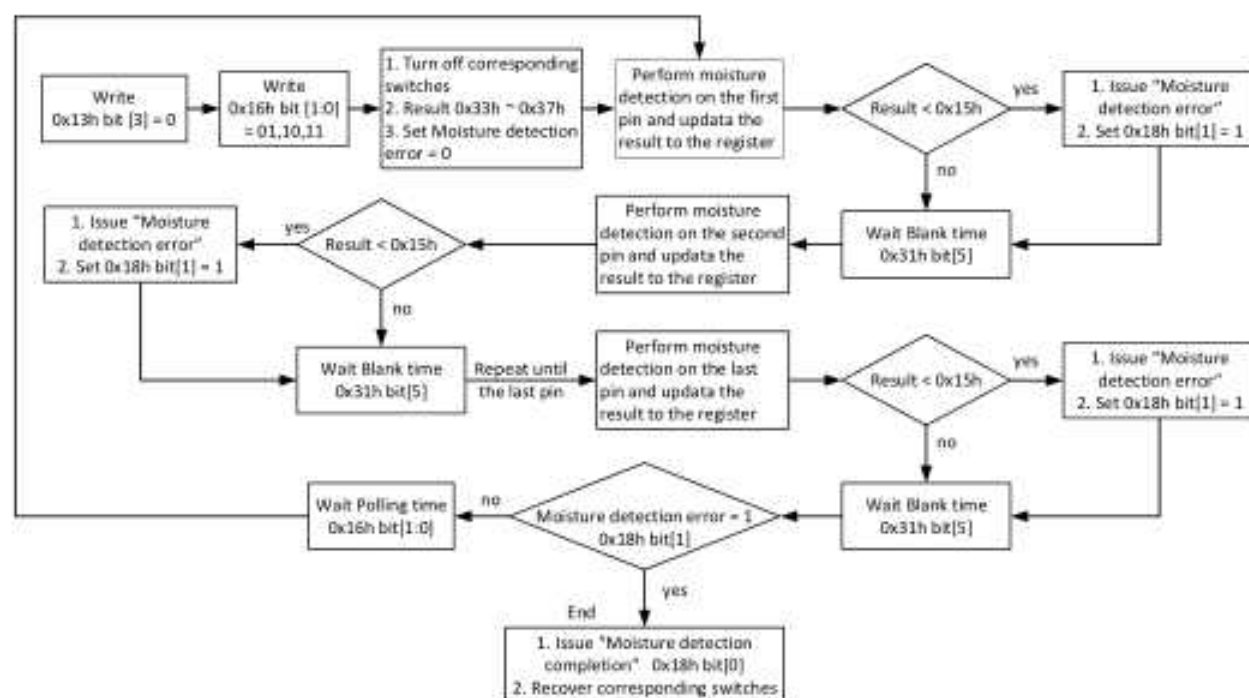


Fig. 12. Multi-pin polling mode (with current source enabled)

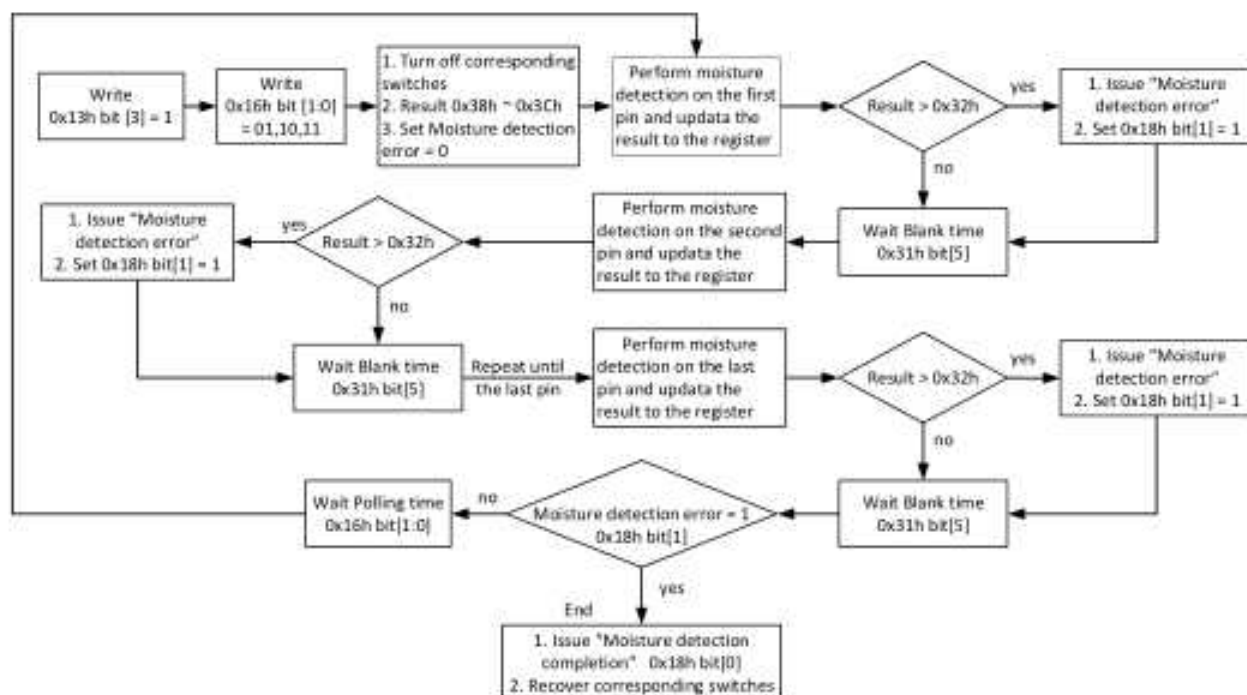


Fig. 13. Multi-pin polling mode (with current source disabled)

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## 12. Register Maps

| ADDR | Register Name                              | Type | Reset Value | Bit7                                                                     | Bit6                    | Bit5                                    | Bit4                      | Bit3                                  | Bit2                                                           | Bit1                                                    | Bit0                                        |  |
|------|--------------------------------------------|------|-------------|--------------------------------------------------------------------------|-------------------------|-----------------------------------------|---------------------------|---------------------------------------|----------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------|--|
| 00H  | Device ID                                  | R    | 0xF6        | 1                                                                        | 1                       | 1                                       | 1                         | 0                                     | 1                                                              | 0                                                       | 1                                           |  |
| 02H  | OVP interrupt flag                         | R/C  | 0x00        | Reserved                                                                 | Reserved                | OVP/DP_R                                | OVP/DN_L                  | OVP/SBU1, SBU2[3:2]                   |                                                                | OVP/GSBU1                                               | OVP/GSBU2                                   |  |
| 03H  | OVP status                                 | R    | 0x00        | Reserved [7:6]                                                           |                         | OVP/DP_R                                | OVP/DN_L                  | OVP/SBU1, SBU2[3:2]                   |                                                                | OVP/GSBU1                                               | OVP/GSBU2                                   |  |
| 04H  | Switch settings enable                     | R/W  | 0x98        | Device enable                                                            | SBU1_H to SBUx switches | SBU2_H to SBUx switches                 | DN_L to DN or L switches  | DP_R to DP or R switches              | Sense to GSBUx switches                                        | MIC to SBUx switches                                    | AGND to SBUx switches                       |  |
| 05H  | Switch select                              | R/W  | 0x18        | USB2 switch select                                                       | SBU1_H switches         | SBU2_H switches                         | DN_L to DN1 or L switches | DP_R to DP1 or R switches             | Sense to GSBUx switches                                        | MIC to SBUx switches                                    | AGND to SBUx switches                       |  |
| 06H  | Switch status0                             | R    | 0x05        | Reserved [7:6]                                                           |                         | Sense switch status [5:4]               |                           | DP_R switch status [3:2]              |                                                                | DN_L switch status [1:0]                                |                                             |  |
| 07H  | Switch status1                             | R    | 0x00        | Reserved [7:6]                                                           |                         | SBU2 switch status [5:3]                |                           |                                       | SBU1 switch status [2:0]                                       |                                                         |                                             |  |
| 08H  | Audio switch left channel turn-on control  | R/W  | 0x01        | Audio switch left channel slow control                                   |                         |                                         |                           |                                       |                                                                |                                                         |                                             |  |
| 09H  | Audio switch right channel turn-on control | R/W  | 0x01        | Audio switch right channel slow control                                  |                         |                                         |                           |                                       |                                                                |                                                         |                                             |  |
| 0AH  | MIC switch turn-on control                 | R/W  | 0x01        | MIC switch slow control                                                  |                         |                                         |                           |                                       |                                                                |                                                         |                                             |  |
| 0BH  | Sense switch turn-on control               | R/W  | 0x01        | Sense switch slow control                                                |                         |                                         |                           |                                       |                                                                |                                                         |                                             |  |
| 0CH  | Audio ground switch turn-on control        | R/W  | 0x01        | Audio ground switch slow control                                         |                         |                                         |                           |                                       |                                                                |                                                         |                                             |  |
| 11H  | Audio accessory status                     | R    | 0x02        | Reserved [7:2]                                                           |                         |                                         |                           |                                       |                                                                | CC_IN                                                   | Reserved                                    |  |
| 12H  | Function enable                            | R/W  | 0x08        | OVP threshold voltage configuration [7:6]                                |                         | Current source for resistance detection | Role of Pin C2 and C3     | Slow turn on control enable           | MIC auto break out control enable                              | Moisture detection enable                               | Audio jack detection & configuration enable |  |
| 13H  | Detection control0                         | R/W  | 0x00        | Moisture detection force off control                                     | Reserved [6:4]          |                                         |                           | Current source for moisture detection | Select the pin for moisture detection in single-pin mode [2:0] |                                                         |                                             |  |
| 14H  | Single-pin mode resistor detection results | R    | 0xFF        | Moisture detection result in single-pin mode with current source enabled |                         |                                         |                           |                                       |                                                                |                                                         |                                             |  |
| 15H  | Moisture resistor detection threshold      | R/W  | 0x16        | Threshold moisture detection with current source enabled                 |                         |                                         |                           |                                       |                                                                |                                                         |                                             |  |
| 16H  | Function control1                          | R/W  | 0x00        | Reserved [7:3]                                                           |                         |                                         |                           |                                       | Select moisture detection mode                                 | Set the time span between two moisture detections [1:0] |                                             |  |
| 17H  | MIC detection status                       | R    | 0x01        | Reserved [7:4]                                                           |                         |                                         |                           | 4 pole                                | 4 pole                                                         | 3 pole                                                  | No audio                                    |  |

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## USB Type-C Analog Audio Switch with Protection Function, Moisture and Audio Jack Detection

|     |                                                   |     |      |                                                                           |                |                                              |                       |                                           |                               |                                    |                   |           |
|-----|---------------------------------------------------|-----|------|---------------------------------------------------------------------------|----------------|----------------------------------------------|-----------------------|-------------------------------------------|-------------------------------|------------------------------------|-------------------|-----------|
| 18H | System flag                                       | R/C | 0x00 | Reserved [7:4]                                                            |                |                                              | Watchdog timeout      | Audio jack detection & configuration      | Moisture detection error      | Moisture detection completion      |                   |           |
| 19H | System mask                                       | R/W | 0x00 | Reserved [7:4]                                                            |                |                                              | Watchdog timeout mask | Audio jack detection & configuration mask | Moisture detection error mask | Moisture detection completion mask |                   |           |
| 1CH | MIC detection threshold data0                     | R/W | 0x20 | MIC detection threshold data                                              |                |                                              |                       |                                           |                               |                                    |                   |           |
| 1EH | I2C reset                                         | W/C | 0x00 | Reserved [7:1]                                                            |                |                                              |                       |                                           |                               |                                    |                   | I2C reset |
| 1FH | Bias current setting                              | R/W | 0x07 | Reserved [7:3]                                                            |                |                                              |                       | Bias current setting [2:0]                |                               |                                    |                   |           |
| 20H | Watchdog setting                                  | R/W | 0x03 | Watchdog enable                                                           | Reserved [6:3] |                                              |                       | Watchdog timer [2:0]                      |                               |                                    |                   |           |
| 30H | Detection data1                                   | R   | 0x00 | Moisture detection result in single-pin mode with current source disabled |                |                                              |                       |                                           |                               |                                    |                   |           |
| 31H | Detection control1                                | R/W | 0x00 | Reserved [7:6]                                                            |                | Multi-pin mode moisture detection blank time | Select pin (CC_IN)    | Select pin (DP_R)                         | Select pin (DN_L)             | Select pin (SBU1)                  | Select pin (SBU2) |           |
| 32H | Moisture voltage detection threshold              | R/W | 0x00 | Threshold for moisture detection with current source disabled             |                |                                              |                       |                                           |                               |                                    |                   |           |
| 33H | Multi-pin mode resistor detection result of CC_IN | R   | 0xFF | Moisture detection result in multi-pin mode with current source enabled   |                |                                              |                       |                                           |                               |                                    |                   |           |
| 34H | Multi-pin mode resistor detection result of DP_R  | R   | 0xFF | Moisture detection result in multi-pin mode with current source enabled   |                |                                              |                       |                                           |                               |                                    |                   |           |
| 35H | Multi-pin mode resistor detection result of DN_L  | R   | 0xFF | Moisture detection result in multi-pin mode with current source enabled   |                |                                              |                       |                                           |                               |                                    |                   |           |
| 36H | Multi-pin mode resistor detection result of SBU1  | R   | 0xFF | Moisture detection result in multi-pin mode with current source enabled   |                |                                              |                       |                                           |                               |                                    |                   |           |
| 37H | Multi-pin mode resistor detection result of SBU2  | R   | 0xFF | Moisture detection result in multi-pin mode with current source enabled   |                |                                              |                       |                                           |                               |                                    |                   |           |
| 38H | Multi-pin mode voltage detection result of CC_IN  | R   | 0x00 | Moisture detection result in multi-pin mode with current source disabled  |                |                                              |                       |                                           |                               |                                    |                   |           |
| 39H | Multi-pin mode voltage detection result of DP_R   | R   | 0x00 | Moisture detection result in multi-pin mode with current source disabled  |                |                                              |                       |                                           |                               |                                    |                   |           |
| 3AH | Multi-pin mode voltage detection result of DN_L   | R   | 0x00 | Moisture detection result in multi-pin mode with current source disabled  |                |                                              |                       |                                           |                               |                                    |                   |           |
| 3BH | Multi-pin mode voltage detection result of SBU1   | R   | 0x00 | Moisture detection result in multi-pin mode with current source disabled  |                |                                              |                       |                                           |                               |                                    |                   |           |
| 3CH | Multi-pin mode voltage detection result of SBU2   | R   | 0x00 | Moisture detection result in multi-pin mode with current source disabled  |                |                                              |                       |                                           |                               |                                    |                   |           |

## EMS4485

### USB Type-C Analog Audio Switch with Protection Function, Moisture and Audio Jack Detection

#### 12.1. I<sup>2</sup>C slave address

| Bit7 | Bit6 | Bit5 | Bit4 | Bit3 | Bit2 | Bit1 | Bit0 |
|------|------|------|------|------|------|------|------|
| 1    | 0    | 0    | 0    | 0    | 1    | 0    | R/W  |

#### 12.2. Register definition

##### 12.2.1. Device ID

Address: 00h

Reset Value: 8'b 1111\_0110

Type: Read

| Bits  | Name        | Size | Description         |
|-------|-------------|------|---------------------|
| [7:6] | Vendor ID   | 2    | Vendor ID           |
| [5:3] | Version ID  | 3    | Device version ID   |
| [2:0] | Revision ID | 3    | Revision history ID |

##### 12.2.2. OVP interrupt flag

Address: 02h

Reset Value: 8'b 0000\_0000

Type: Read Clear

| Bits  | Name             | Size | Description                                                                             |
|-------|------------------|------|-----------------------------------------------------------------------------------------|
| [7:6] | Reserved         | 2    | Do not use                                                                              |
| 5     | DP_R OVP         | 1    | 0: OVP event has not occurred<br>1: OVP event has occurred                              |
| 4     | DN_L OVP         | 1    | 0: OVP event has not occurred<br>1: OVP event has occurred                              |
| [3:2] | SBU1    SBU2 OVP | 2    | 0: OVP event has not occurred<br>1: At least one of SBU1 or SBU2 OVP event has occurred |
| 1     | GSRU1 OVP        | 1    | 0: OVP event has not occurred<br>1: OVP event has occurred                              |
| 0     | GSRU2 OVP        | 1    | 0: OVP event has not occurred<br>1: OVP event has occurred                              |

##### 12.2.3. OVP status

Address: 03h

Reset Value: 8'b 0000\_0000

Type: Read

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### USB Type-C Analog Audio Switch with Protection Function, Moisture and Audio Jack Detection

| Bits  | Name              | Size | Description                                                                             |
|-------|-------------------|------|-----------------------------------------------------------------------------------------|
| [7:6] | Reserved          | 2    | Do not use                                                                              |
| 5     | OVP on DP_R PIN   | 1    | 0: OVP event has not occurred<br>1: OVP event has occurred                              |
| 4     | OVP on DN_L PIN   | 1    | 0: OVP event has not occurred<br>1: OVP event has occurred                              |
| [3:2] | SBU1    SBU2 OVP  | 2    | 0: OVP event has not occurred<br>1: At least one of SBU1 or SBU2 OVP event has occurred |
| 1     | OVP on GSBUS1 PIN | 1    | 0: OVP event has not occurred<br>1: OVP event has occurred                              |
| 0     | OVP on GSBUS2 PIN | 1    | 0: OVP event has not occurred<br>1: OVP event has occurred                              |

#### 12.2.4. Switching setting enable

Address: 04h

Reset Value: 8'b 1001\_1000

Type: Read/Write

| Bits | Name                        | Size | Description                                                                                                               |
|------|-----------------------------|------|---------------------------------------------------------------------------------------------------------------------------|
| 7    | Device enable               | 1    | 1: Device enable.<br>0: Device disable; L, R pull down by 10 kΩ and other switch nodes will be high-Z for positive input. |
| 6    | SBU1_H to SBUx switches     | 1    | 0: Switch disable; SBU1_H will be pull-down to ground.<br>1: Switch enable                                                |
| 5    | SBU2_H to SBUx switches     | 1    | 0: Switch disable; SBU2_H will be pull-down to ground.<br>1: Switch enable                                                |
| 4    | DN_L to DN1/2 or L switches | 1    | 0: Switch disable; DN_L, DN1/2 will be high-Z for positive input. L pull down by 10 kΩ.<br>1: Switch enable               |
| 3    | DP_R to DP1/2 or R switches | 1    | 0: Switch disable; DP_R, DP1/2 will be high-Z for positive input. R pull down by 10 kΩ.<br>1: Switch enable               |
| 2    | Sense to GSBUSx switches    | 1    | 0: Switch disable; Sense, GSBUS1 and GSBUS2 will be pull-down to ground.<br>1: Switch enable                              |
| 1    | MIC to SBUx switches        | 1    | 0: Switch disable; MIC will be pull-down to ground.<br>1: Switch enable                                                   |
| 0    | AGND to SBUx switches       | 1    | 0: Switch disable; AGND will be high-Z for positive input.<br>1: Switch enable                                            |

#### 12.2.5. Switch select

Address: 05h

Reset Value: 8'b 0001\_1000

Type: Read/Write

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### USB Type-C Analog Audio Switch with Protection Function, Moisture and Audio Jack Detection

| Bits | Name                      | Size | Description                                                                                                                                                             |
|------|---------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | USB2 switch select        | 1    | 0: USB2 off. USB1 or Audio switch depend on 05h, bit [4:3] and 04h, bit [4:3]<br>1: DP/R~DP2, DN/L~DN2 switches ON, if 04h, bit [4:3] = '11', and 05h, bit [4:3] = '11' |
| 6    | SBU1_H switches           | 1    | 0: SBU1_H to SBU1 switch ON<br>1: SBU1_H to SBU2 switch ON                                                                                                              |
| 5    | SBU2_H switches           | 1    | 0: SBU2_H to SBU2 switch ON<br>1: SBU2_H to SBU1 switch ON                                                                                                              |
| 4    | DN_L to DN1 or L switches | 1    | 0: DN_L to L switch ON<br>1: DN_L to DN1 switch ON                                                                                                                      |
| 3    | DP_R to DP1 or R switches | 1    | 0: DP_R to R switch ON<br>1: DP_R to DP1 switch ON                                                                                                                      |
| 2    | Sense to GSBUX switches   | 1    | 0: Sense to GSBU1 switch ON<br>1: Sense to GSBU2 switch ON                                                                                                              |
| 1    | MIC to SBUx switches      | 1    | 0: MIC to GSBU2 switch ON<br>1: MIC to GSBU1 switch ON                                                                                                                  |
| 0    | AGND to SBUx switches     | 1    | 0: AGND to SBU1 switch ON<br>1: AGND to SBU2 switch ON                                                                                                                  |

**Note:** If want to change to USB2 mode, you must first write 0x05h = 0x98h, then write 0x04h = 0x98h

#### 12.2.6. Switch status0

Address: 06h

Reset Value: 8'b 0000\_0101

Type: Read

| Bits  | Name                | Size | Description                                                                                                               |
|-------|---------------------|------|---------------------------------------------------------------------------------------------------------------------------|
| [7:6] | Reserved            | 2    | Do not use                                                                                                                |
| [5:4] | Sense switch status | 2    | 00: Sense switch is Open / Not Connected<br>01: Sense connected to GSBU1<br>10: Sense connected to GSBU2<br>11: Not valid |
| [3:2] | DP_R switch status  | 2    | 00: DP_R Switch Open / Not Connected<br>01: DP_R connected to DP1<br>10: DP_R connected to R<br>11: DP_R connected to DP2 |
| [1:0] | DN_L switch status  | 2    | 00: DN_L Switch Open / Not Connected<br>01: DN_L connected to DN1<br>10: DN_L connected to L<br>11: DN_L connected to DN2 |

#### 12.2.7. Switch status1

Address: 07h

Reset Value: 8'b 0000\_0000

Type: Read

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### USB Type-C Analog Audio Switch with Protection Function, Moisture and Audio Jack Detection

| Bits  | Name                     | Size | Description                                                                                                                                                                                                                                                                        |
|-------|--------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [7:6] | Reserved                 | 2    | Do not use                                                                                                                                                                                                                                                                         |
| [5:3] | SBU2/GSBU2 switch status | 3    | 000: SBU2/GSBU2 switch is Open / Not Connected<br>001: SBU2/GSBU2 connected to MIC<br>010: SBU2/GSBU2 connected to AGND<br>011: SBU2/GSBU2 connected to SBU1_H<br>100: SBU2/GSBU2 connected to SBU2_H<br>101: SBU2/GSBU2 connected both SBU1_H and SBU2_H<br>110...111: Do not use |
| [2:0] | SBU1/GSBU1 switch status | 3    | 000: SBU1/GSBU1 switch is Open / Not Connected<br>001: SBU1/GSBU1 connected to MIC<br>010: SBU1/GSBU1 connected to AGND<br>011: SBU1/GSBU1 connected to SBU1_H<br>100: SBU1/GSBU1 connected to SBU2_H<br>101: SBU1/GSBU1 connected both SBU1_H and SBU2_H<br>110...111: Do not use |

#### 12.2.8. Audio switch left channel slow turn-on

Address: 08h

Reset Value: 8'b 0000\_0001

Type: Read/Write

| Bits  | Name                               | Size | Description                                                                                                                                            |
|-------|------------------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| [7:0] | Switch turn on rising time setting | 8    | 11111111: 8300 $\mu$ s<br>...<br>10111111: 645 $\mu$ s<br>...<br>01111111: 327 $\mu$ s<br>...<br>00111111: 222 $\mu$ s<br>...<br>00000001: 114 $\mu$ s |

#### 12.2.9. Audio switch right channel slow turn-on

Address: 09h

Reset Value: 8'b 0000\_0001

Type: Read/Write

| Bits  | Name                               | Size | Description                                                                                                                                            |
|-------|------------------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| [7:0] | Switch turn on rising time setting | 8    | 11111111: 8300 $\mu$ s<br>...<br>10111111: 645 $\mu$ s<br>...<br>01111111: 327 $\mu$ s<br>...<br>00111111: 222 $\mu$ s<br>...<br>00000001: 114 $\mu$ s |

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### USB Type-C Analog Audio Switch with Protection Function, Moisture and Audio Jack Detection

#### 12.2.10. MIC switch slow turn-on

Address: 0Ah

Reset Value: 8'b 0000\_0001

Type: Read/Write

| Bits  | Name                               | Size | Description                                                                                                                                                 |
|-------|------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [7:0] | Switch turn on rising time setting | 8    | 11111111: 57567 $\mu$ s<br>...<br>10111111: 7680 $\mu$ s<br>...<br>01111111: 4833 $\mu$ s<br>...<br>00111111: 3883 $\mu$ s<br>...<br>00000001: 2067 $\mu$ s |

#### 12.2.11. Sense switch slow turn-on

Address: 0Bh

Reset Value: 8'b 0000\_0001

Type: Read/Write

| Bits  | Name                               | Size | Description                                                                                                                                                   |
|-------|------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [7:0] | Switch turn-on rising time setting | 8    | 11111111: 166000 $\mu$ s<br>...<br>10111111: 10883 $\mu$ s<br>...<br>01111111: 5017 $\mu$ s<br>...<br>00111111: 3303 $\mu$ s<br>...<br>00000001: 1273 $\mu$ s |

#### 12.2.12. Audio ground switch slow turn-on

Address: 0Ch

Reset Value: 8'b 0000\_0001

Type: Read/Write

| Bits  | Name                               | Size | Description                                                                                                                                              |
|-------|------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| [7:0] | Switch turn-on rising time setting | 8    | 11111111: 17500 $\mu$ s<br>...<br>10111111: 1327 $\mu$ s<br>...<br>01111111: 632 $\mu$ s<br>...<br>00111111: 427 $\mu$ s<br>...<br>00000001: 263 $\mu$ s |

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### USB Type-C Analog Audio Switch with Protection Function, Moisture and Audio Jack Detection

#### 12.2.13. Audio accessory status

Address: 11h

Reset Value: 8'b 0000\_0010

Type: Read

| Bits  | Name     | Size | Description                          |
|-------|----------|------|--------------------------------------|
| [7:2] | Reserved | 6    | Do not use                           |
| 1     | CC_IN    | 1    | 0: CC_IN < 1.2 V<br>1: CC_IN > 1.5 V |
| 0     | Reserved | 1    | Do not use                           |

#### 12.2.14. Function enable

Address: 12h

Reset Value: 8'b 0000\_1000

Type: Read/Write

| Bits  | Name                                          | Size | Description                                                                                                                                  |
|-------|-----------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------|
| [7:6] | OVP threshold voltage configuration           | 2    | 00: 4.4 V<br>01: 4.6 V<br>10: 4.8 V<br>11: 5.0 V                                                                                             |
| 5     | Current source for resistance detection       | 1    | If 0× 13h bit [3] = 0:<br>0: 5 $\mu$ A<br>1: 1 $\mu$ A                                                                                       |
| 4     | Role of pin C2 and C3                         | 1    | 0: Set C2 as DET and C3 as INT<br>1: Set C2 as COMPIN and C3 as COMPOUT                                                                      |
| 3     | Slow turn on control enable                   | 1    | 1: Enable<br>0: Disable                                                                                                                      |
| 2     | MIC auto break out control enable             | 1    | 1: Enable<br>0: Disable                                                                                                                      |
| 1     | Moisture detection enable                     | 1    | 1: Enable. It would return to '0' after the detection was finished.<br>0: Disable<br>(This bit can be reset by POR, I <sup>2</sup> C or WTD) |
| 0     | Audio jack detection and configuration enable | 1    | 1: Enable. It will be changed to '0' after audio jack detection and configuration<br>0: Disable                                              |

#### 12.2.15. Detection control0

Address: 13h

Reset Value: 8'b 0000\_0000

Type: Read/Write

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### USB Type-C Analog Audio Switch with Protection Function, Moisture and Audio Jack Detection

| Bits  | Name                                                     | Size | Description                                                                                                               |
|-------|----------------------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------|
| 7     | Moisture detection force off control                     | 1    | 0: Moisture detection on and off control would follow 0×12h bit [1]<br>1: Moisture detection would be turned off forcibly |
| [6:4] | Reserved                                                 | 3    | Do not use                                                                                                                |
| 3     | Current source for moisture detection                    | 1    | 0: Enable<br>1: Disable                                                                                                   |
| [2:0] | Select the pin for moisture detection in single-pin mode | 3    | 000: CC_IN<br>001: DP_R<br>010: DN_L<br>011: SBU1<br>100: SBU2<br>101...111: Reserved                                     |

#### 12.2.16. Single-pin mode resistor detection results

Address: 14h

Reset Value: 8'b 1111\_1111

Type: Read

| Bits  | Name                                                                     | Size | Description                                                                                   |
|-------|--------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------|
| [7:0] | Moisture detection result in single-pin mode with current source enabled | 8    | 0x00 ~ 0xFF:<br>2 kΩ * Bits, with 0×12h bit [5] = 0;<br>10 kΩ * Bits, with 0×12h bit [5] = 1. |

#### 12.2.17. Moisture resistor detection threshold

Address: 15h

Reset Value: 8'b 0001\_0110

Type: Read/Write

| Bits  | Name                                                          | Size | Description                                                                                                                                                 |
|-------|---------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [7:0] | Threshold for moisture detection with current source enabled. | 8    | Would issue 'Moisture detection error' once any contents from 0×14h (in single pin mode) or 0×33h ~ 0×37h (in multi-pin mode) is lower than this threshold. |

#### 12.2.18. Function control1

Address: 16h

Reset Value: 8'b 0000\_0000

Type: Read/Write

| Bits  | Name                           | Size | Description                                                                                                            |
|-------|--------------------------------|------|------------------------------------------------------------------------------------------------------------------------|
| [7:3] | Reserved                       | 5    | Do not use.                                                                                                            |
| 2     | Select moisture detection mode | 1    | 0: Single-pin mode, will perform moisture on single pin.<br>1: Multi-pin mode, will perform moisture on multiple pins. |

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### USB Type-C Analog Audio Switch with Protection Function, Moisture and Audio Jack Detection

|       |                                                   |   |                                                                                                                                                                                                                                                                                                         |
|-------|---------------------------------------------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [1:0] | Set the time span between two moisture detections | 2 | 00: No polling.<br>In single-pin mode, perform single detection for the selected pin (0x13h bit [2:0]).<br>In multi-pin mode, perform single detection for all selected pins (0x31h bit [4:0]).<br>01: Polling with 400 ms time span<br>10: Polling with 2s time span<br>11: Polling with 20s time span |
|-------|---------------------------------------------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 12.2.19. MIC detection status

Address: 17h

Reset Value: 8'b 0000\_0001

Type: Read

| Bits  | Name               | Size | Description                                              |
|-------|--------------------|------|----------------------------------------------------------|
| [7:4] | Reserved           | 4    | Do not use                                               |
| 3     | 4 pole             | 1    | 1: 4 Pole SBU2 to MIC, SBU1 to audio ground<br>0: others |
| 2     | 4 pole             | 1    | 1: 4 Pole SBU1 to MIC, SBU2 to audio ground<br>0: others |
| 1     | 3 pole             | 1    | 1: 3 pole<br>0: others                                   |
| 0     | No audio accessory | 1    | 1: No audio accessory<br>0: Audio accessory attached     |

#### 12.2.20. System flag

Address: 18h

Reset Value: 8'b 0000\_0000

Type: Read Clear

| Bits  | Name                                   | Size | Description                                                                                                                                                                                                                                   |
|-------|----------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [7:4] | Reserved                               | 4    | Do not use                                                                                                                                                                                                                                    |
| 3     | Watchdog timeout                       | 1    | 0: Watchdog timeout has not occurred<br>1: Watchdog timeout has occurred                                                                                                                                                                      |
| 2     | Audio jack detection and configuration | 1    | 0: Audio jack detection and configuration have not completed.<br>1: Audio jack detection and configuration is completed.                                                                                                                      |
| 1     | Moisture detection error               | 1    | 0: Moisture doesn't occur.<br>For resistance detection, result(s) $\geq$ TH0<br>For voltage detection, result(s) $\leq$ TH1<br>1: Moisture occurs.<br>For resistance detection, result(s) $<$ TH0<br>For voltage detection, result(s) $>$ TH1 |
| 0     | Moisture detection completion          | 1    | 0: Moisture detection has been not completed<br>1: Moisture detection is completed.                                                                                                                                                           |

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### USB Type-C Analog Audio Switch with Protection Function, Moisture and Audio Jack Detection

#### 12.2.21. System mask

Address: 19h

Reset Value: 8'b 0000\_0000

Type: Read/Write

| Bits  | Name                                        | Size | Description                                                                                                                  |
|-------|---------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------|
| [7:4] | Reserved                                    | 4    | Do not use.                                                                                                                  |
| 3     | Watchdog timeout mask                       | 1    | 0: "Watchdog timeout" bit is not masked.<br>1: "Watchdog timeout" bit is masked.                                             |
| 2     | Audio jack detection and configuration mask | 1    | 0: "Audio jack detection and configuration" bit is not masked.<br>1: "Audio jack detection and configuration" bit is masked. |
| 1     | Moisture detection error mask               | 1    | 0: "Moisture detection error" bit is not masked.<br>1: "Moisture detection error" bit is masked.                             |
| 0     | Moisture detection completion mask          | 1    | 0: "Moisture detection completion" bit is not masked.<br>1: "Moisture detection completion" bit is masked.                   |

#### 12.2.22. MIC detection threshold data

Address: 1Ch

Reset Value: 8'b 0010\_0000

Type: Write

| Bits  | Name                         | Size | Description                   |
|-------|------------------------------|------|-------------------------------|
| [7:0] | MIC detection threshold data | 8    | MIC detection threshold data. |

#### 12.2.23. I2C reset

Address: 1Eh

Reset Value: 8'b 0000\_0000

Type: Write / Clear

| Bits  | Name                   | Size | Description                             |
|-------|------------------------|------|-----------------------------------------|
| [7:1] | Reserved               | 7    | Reserved                                |
| 0     | I <sup>2</sup> C reset | 1    | 0: default<br>1: I <sup>2</sup> C reset |

#### 12.2.24. Bias current setting

Address: 1Fh

Reset Value: 8'b 0000\_0111

Type: Read/Write

| Bits  | Name     | Size | Description |
|-------|----------|------|-------------|
| [7:3] | Reserved | 4    | Do not use  |

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### USB Type-C Analog Audio Switch with Protection Function, Moisture and Audio Jack Detection

|       |                      |   |                                                                                          |
|-------|----------------------|---|------------------------------------------------------------------------------------------|
| [2:0] | Bias current setting | 4 | Setting bias current for audio jack detection.<br>0000 ~ 1111: Bits $\times$ 100 $\mu$ A |
|-------|----------------------|---|------------------------------------------------------------------------------------------|

#### 12.2.25. Watchdog setting

Address: 20h

Reset Value: 8'b 0000\_0011

Type: Read/Write

| Bits  | Name            | Size | Description                                                                                         |
|-------|-----------------|------|-----------------------------------------------------------------------------------------------------|
| 7     | Watchdog enable | 1    | 0: Watchdog disabled<br>1: Watchdog enabled                                                         |
| [6:3] | Reserved        | 4    | Do not use                                                                                          |
| [2:0] | Watchdog timer  | 3    | 000: 0.5 s<br>001: 1 s<br>010: 2 s<br>011: 5 s<br>100: 10 s<br>101: 30 s<br>110: 60 s<br>111: 5 min |

#### 12.2.26. Detection data1

Address: 30h

Reset Value: 8'b 0000\_0000

Type: Read

| Bits  | Name            | Size | Description                                                                                                                         |
|-------|-----------------|------|-------------------------------------------------------------------------------------------------------------------------------------|
| [7:0] | Detection data1 | 8    | Moisture detection result in single-pin mode with current source disabled:<br>0x00 ~ 0xFF: Bits $\times$ 9 $\times$ 0.001 [Unit: V] |

#### 12.2.27. Detection control1

Address: 31h

Reset Value: 8'b 0000\_0000

Type: Read/Write

| Bits  | Name                                         | Size | Description                                                                                       |
|-------|----------------------------------------------|------|---------------------------------------------------------------------------------------------------|
| [7:6] | Reserved                                     | 2    | Do not use.                                                                                       |
| 5     | Multi-pin mode moisture detection blank time | 1    | In multi-pin mode, set blank time between detections for different pins.<br>0: 20 ms<br>1: 120 ms |
| 4     | Select pin (CC_IN)                           | 1    | Select pin for moisture detection in multi-pin mode.<br>0: not selected<br>1: selected            |
| 3     | Select pin (DP_R)                            | 1    | Select pin for moisture detection in multi-pin mode.<br>0: not selected<br>1: selected            |

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### USB Type-C Analog Audio Switch with Protection Function, Moisture and Audio Jack Detection

|   |                   |   |                                                                                        |
|---|-------------------|---|----------------------------------------------------------------------------------------|
| 2 | Select pin (DN_L) | 1 | Select pin for moisture detection in multi-pin mode.<br>0: not selected<br>1: selected |
| 1 | Select pin (SBU1) | 1 | Select pin for moisture detection in multi-pin mode.<br>0: not selected<br>1: selected |
| 0 | Select pin (SBU2) | 1 | Select pin for moisture detection in multi-pin mode.<br>0: not selected<br>1: selected |

#### 12.2.28. Moisture voltage detection threshold

Address: 32h

Reset Value: 8'b 0000\_0000

Type: Read/Write

| Bits  | Name                                                          | Size | Description                                                                                                                                                                                                                               |
|-------|---------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [7:0] | Threshold for moisture detection with current source disabled | 8    | Threshold for moisture detection with current source disabled. EMS4485 would issue "Moisture detection error" as soon as any contents from 0x30h (in single-pin mode) or 0x38h ~ 0x3Ch (in multi-pin mode) is higher than this threshold. |

#### 12.2.29. Multi-pin mode resistor detection result of CC\_IN

Address: 33h

Reset Value: 8'b 1111\_1111

Type: Read

| Bits  | Name                                         | Size | Description                                                                                                                                                                                 |
|-------|----------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [7:0] | Moisture detection in multi-pin mode (CC_IN) | 8    | Moisture detection result in multi-pin mode with current source enabled:<br>0x00 ~ 0xFF:<br>2 k $\Omega$ * Bits, with 0x 12h bit [5] = 0;<br>10 k $\Omega$ * Bits, with 0x 12h bit [5] = 1. |

#### 12.2.30. Multi-pin mode resistor detection result of DP\_R

Address: 34h

Reset Value: 8'b 1111\_1111

Type: Read

| Bits  | Name                                        | Size | Description                                                                                                                                                                                 |
|-------|---------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [7:0] | Moisture detection in multi-pin mode (DP_R) | 8    | Moisture detection result in multi-pin mode with current source enabled:<br>0x00 ~ 0xFF:<br>2 k $\Omega$ * Bits, with 0x 12h bit [5] = 0;<br>10 k $\Omega$ * Bits, with 0x 12h bit [5] = 1. |

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### USB Type-C Analog Audio Switch with Protection Function, Moisture and Audio Jack Detection

#### 12.2.31. Multi-pin mode resistor detection result of DN\_L

Address: 35h

Reset Value: 8'b 1111\_1111

Type: Read

| Bits  | Name                                        | Size | Description                                                                                                                                                                                 |
|-------|---------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [7:0] | Moisture detection in multi-pin mode (DN_L) | 8    | Moisture detection result in multi-pin mode with current source enabled.<br>0x00 ~ 0xFF:<br>2 k $\Omega$ * Bits, with 0x 12h bit [5] = 0;<br>10 k $\Omega$ * Bits, with 0x 12h bit [5] = 1. |

#### 12.2.32. Multi-pin mode resistor detection result of SBU1

Address: 36h

Reset Value: 8'b 1111\_1111

Type: Read

| Bits  | Name                                        | Size | Description                                                                                                                                                                                 |
|-------|---------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [7:0] | Moisture detection in multi-pin mode (SBU1) | 8    | Moisture detection result in multi-pin mode with current source enabled.<br>0x00 ~ 0xFF:<br>2 k $\Omega$ * Bits, with 0x 12h bit [5] = 0;<br>10 k $\Omega$ * Bits, with 0x 12h bit [5] = 1. |

#### 12.2.33. Multi-pin mode resistor detection result of SBU2

Address: 37h

Reset Value: 8'b 1111\_1111

Type: Read

| Bits  | Name                                        | Size | Description                                                                                                                                                                                 |
|-------|---------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [7:0] | Moisture detection in multi-pin mode (SBU2) | 8    | Moisture detection result in multi-pin mode with current source enabled.<br>0x00 ~ 0xFF:<br>2 k $\Omega$ * Bits, with 0x 12h bit [5] = 0;<br>10 k $\Omega$ * Bits, with 0x 12h bit [5] = 1. |

#### 12.2.34. Multi-pin mode voltage detection result of CC\_IN

Address: 38h

Reset Value: 8'b 0000\_0000

Type: Read

| Bits  | Name                                         | Size | Description                                                                                                                        |
|-------|----------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------|
| [7:0] | Moisture detection in multi-pin mode (CC_IN) | 8    | Moisture detection result in multi-pin mode with current source disabled.<br>0x00 ~ 0xFF: Bits $\times$ 9 $\times$ 0.001 [unit: V] |

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#### 12.2.35. Multi-pin mode voltage detection result of DP\_R

Address: 39h

Reset Value: 8'b 0000\_0000

Type: Read

| Bits  | Name                                        | Size | Description                                                                                                                   |
|-------|---------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------|
| [7:0] | Moisture detection in multi-pin mode (DP_R) | 8    | Moisture detection result in multi-pin mode with current source disabled. 0x00 ~ 0xFF: Bits $\times 9 \times 0.001$ [unit: V] |

#### 12.2.36. Multi-pin mode voltage detection result of DN\_L

Address: 3Ah

Reset Value: 8'b 0000\_0000

Type: Read

| Bits  | Name                                        | Size | Description                                                                                                                   |
|-------|---------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------|
| [7:0] | Moisture detection in multi-pin mode (DN_L) | 8    | Moisture detection result in multi-pin mode with current source disabled. 0x00 ~ 0xFF: Bits $\times 9 \times 0.001$ [unit: V] |

#### 12.2.37. Multi-pin mode voltage detection result of SBU1

Address: 3Bh

Reset Value: 8'b 0000\_0000

Type: Read

| Bits  | Name                                        | Size | Description                                                                                                                   |
|-------|---------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------|
| [7:0] | Moisture detection in multi-pin mode (SBU1) | 8    | Moisture detection result in multi-pin mode with current source disabled. 0x00 ~ 0xFF: Bits $\times 9 \times 0.001$ [unit: V] |

#### 12.2.38. Multi-pin mode voltage detection result of SBU2

Address: 3Ch

Reset Value: 8'b 0000\_0000

Type: Read

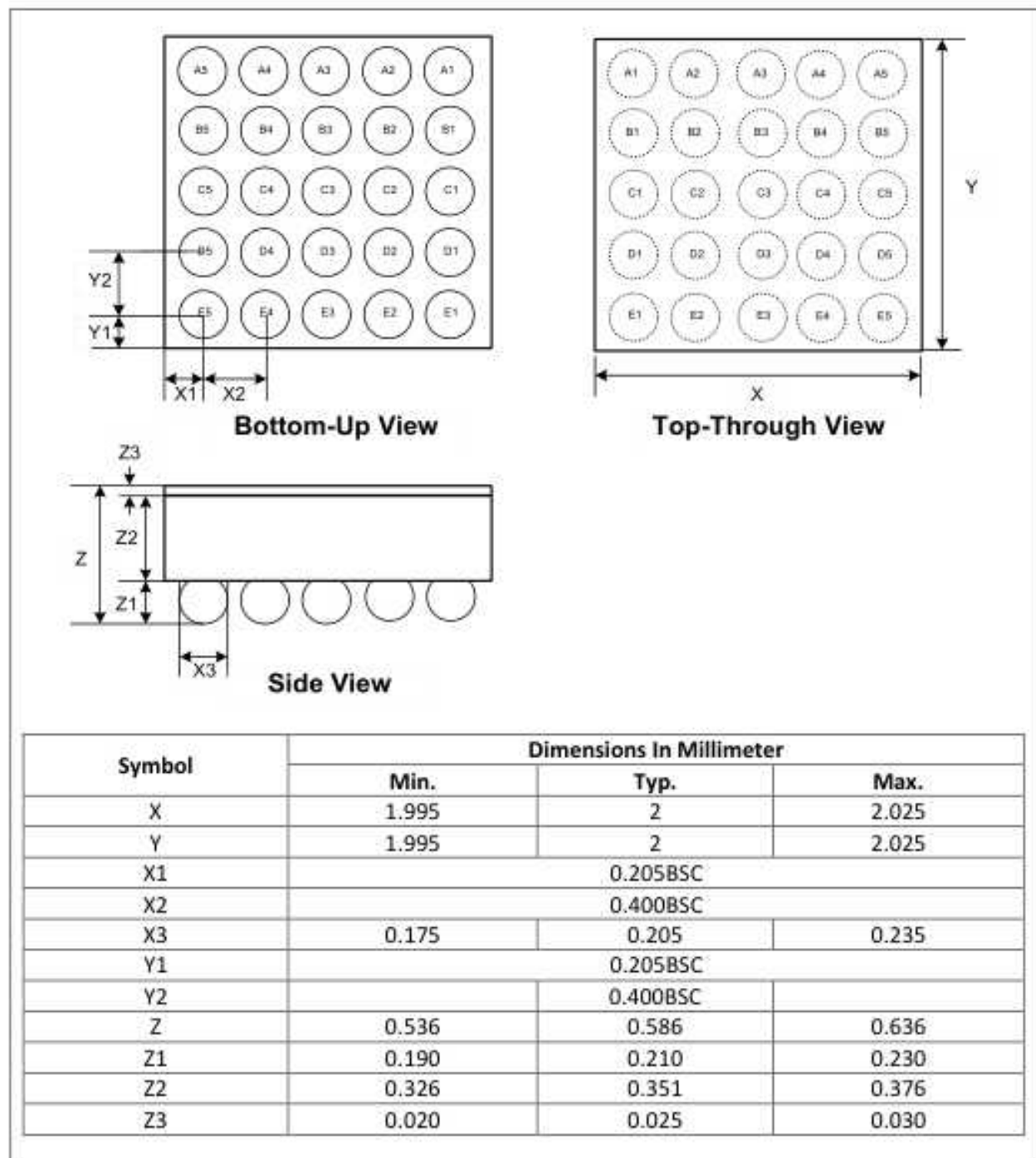
| Bits  | Name                                        | Size | Description                                                                                                                   |
|-------|---------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------|
| [7:0] | Moisture detection in multi-pin mode (SBU2) | 8    | Moisture detection result in multi-pin mode with current source disabled. 0x00 ~ 0xFF: Bits $\times 9 \times 0.001$ [unit: V] |

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## 13. Package Outline

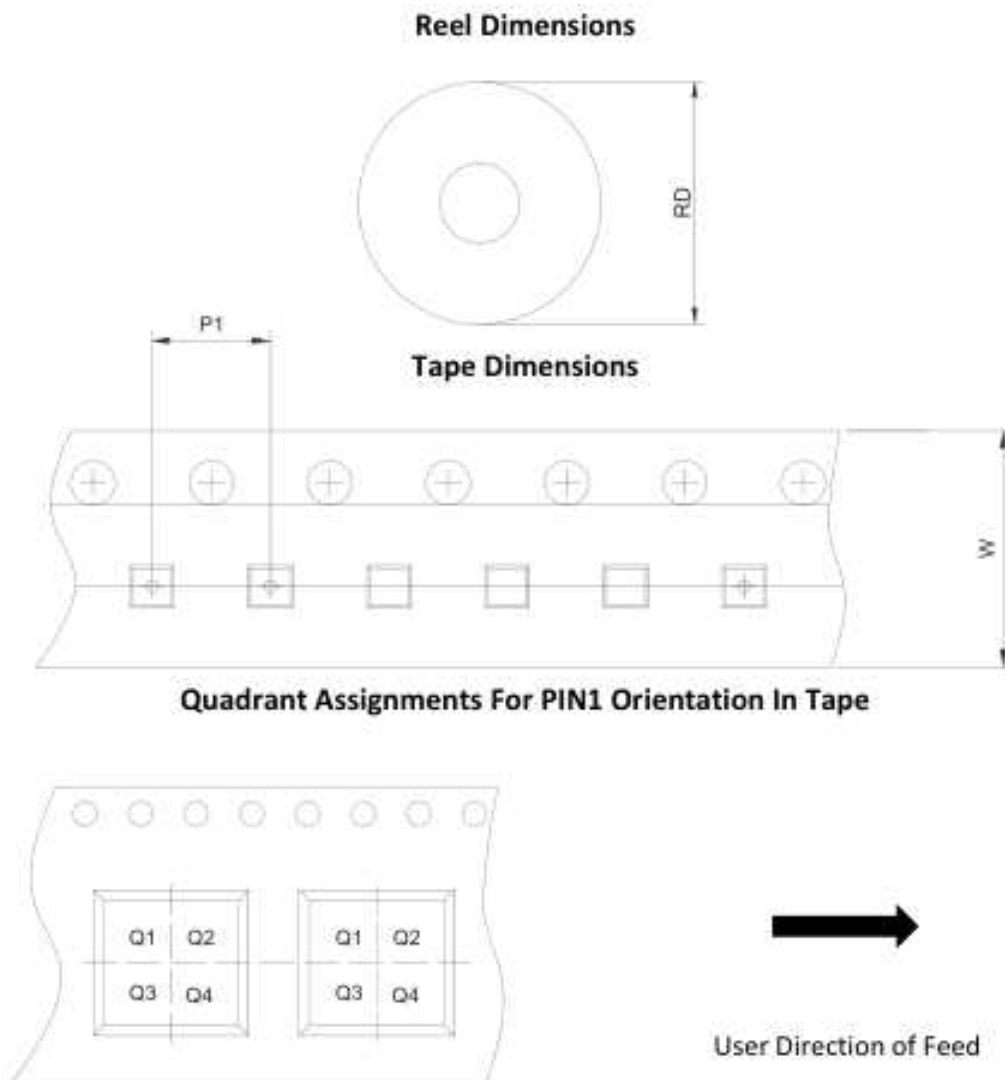
WLCSP-25



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## 14. Tape and Reel Information



|             |                                                |                                                                                                                            |
|-------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>RD</b>   | <b>Reel Dimension</b>                          | <input checked="" type="checkbox"/> 7inch <input type="checkbox"/> 13inch                                                  |
| <b>W</b>    | <b>Overall width of the carrier tape</b>       | <input checked="" type="checkbox"/> 8mm <input type="checkbox"/> 12mm <input type="checkbox"/> 16mm                        |
| <b>P1</b>   | <b>Pitch between successive cavity centers</b> | <input type="checkbox"/> 2mm <input checked="" type="checkbox"/> 4mm <input type="checkbox"/> 8mm                          |
| <b>Pin1</b> | <b>Pin1 Quadrant</b>                           | <input checked="" type="checkbox"/> Q1 <input type="checkbox"/> Q2 <input type="checkbox"/> Q3 <input type="checkbox"/> Q4 |

## 15. Revision History

**Table 8. Revision history**

| Document ID      | Release Date                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Data sheet status | Change notice | Supersedes       |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------|------------------|
| EMS4485 Rev. 0.9 | Jul 06, 2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Draft datasheet   |               | EMS4485 Rev. 0.1 |
| Modifications:   | <ul style="list-style-type: none"> <li>Section 3: Changed the package name from WLCSP-25B to WLCSP-25</li> <li>Section 4: Added pin connection notes for AGND/GND, SBU1/GSBU1, and SBU2/GSBU2.</li> <li>Section 7: Updated Absolute Maximum Ratings: <math>V_{SW\_C}</math>, <math>V_{SW\_Audio}</math>, <math>V_{SW\_USB}</math>, <math>V_{SW\_UART}</math>, <math>V_{VSBUX\_H}</math>, <math>V_{I/O}</math>, <math>V_{CTRL}</math>, <math>V_{comparator}</math>, <math>I_{SW\_UART}</math> updated.</li> <li>Section 8: Updated Recommended Operating Conditions: <math>V_{CC}</math>, <math>V_{SW\_Audio}</math>, <math>V_{SW\_UART}</math>, <math>V_{VSBUX}</math>, <math>V_{IH}</math> updated.</li> <li>Section 9: Updated ESD Ratings</li> <li>Section 10: Updated DC electrical characteristics (Test condition <math>V_{CC}</math> max changed from 5.5V to 4.5V): <math>I_{ON(Audio)}</math>, <math>I_{ON(UART)}</math> updated. <math>R_{FLAT}</math> removed. <math>R_{OFF(Audio)}</math> added.</li> <li>Section 10: Updated AC electrical characteristics (Test condition <math>V_{CC}</math> max changed from 5.5V to 4.5V): Audio switch <math>t_{delay}</math>, Audio switch <math>t_{rise}</math>, USB switch <math>t_{ON/tOFF}</math>, MIC/AGND switch timing updated. UART switch <math>t_{ON/tOFF}</math> added.</li> <li>Section 10: Added 10.3 Comparator electrical characteristics and 10.4 Comparator electrical characteristics</li> <li>Section 12: Updated Register Maps<br/>           0x04h: Disabled state behavior changed from high-Z to pull-down to ground (Bits 6, 5, 2, 1).<br/>           0x07h: Register names updated to SBU2/GSBU2 status (Bits [5:3]) and SBU1/GSBU1 status (Bits [2:0]).<br/>           0x08h~0x0Ch: R/L Switch, MIC switch, Sense switch, Audio ground switch turn-on rising time ranges updated.<br/>           0x16h: Polling time span options updated.<br/>           0x31h: Multi-pin mode blank time options updated.</li> </ul> |                   |               |                  |
| EMS4485 Rev. 0.1 | May 06, 2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Draft datasheet   |               |                  |